

Title

Abstract

We developed a unified non-parametric framework for change-point detection in multivariate, locally stationary time series defined via nonlinear moment equations $\mathbb{E}[H(X_{t,n}, \theta_{t,n})] = 0$. The proposed methodology encompasses a broad range of statistical targets, including robust location, time-varying M - regression with redescending score functions, autocovariances, and smoothed extremal downside correlation (SEDC) densities. To circumvent the slow convergence rates and boundary bias inherent to standard plug-in estimators, we construct a linearized partial-sum estimator based on a local one-step Taylor expansion with an independent decoupling lag. Under mild physical dependence conditions and smooth kernel assumptions, we establish uniform consistency of the local pilot estimators and prove a Functional Central Limit Theorem (FCLT) in the Skorokhod space, showing weak convergence of the linearized process toward a Gaussian martingale driven by the local long-run covariance structure. Furthermore, we prove the validity of a multiplier block bootstrap to approximate the asymptotic null distribution of the resulting CUSUM test statistic. Extensive Monte Carlo simulations demonstrate that the proposed linearized CUSUM test maintains accurate nominal size under severe nuisance non-stationarity and heavy-tailed innovations, while achieving substantial power gains and bias reduction over classical baselines under additive, innovation, and replacement outlier regimes.

1 Introduction

2 Model

In this paper, we study a multivariate, locally stationary time series model given by the causal representation

$$X_{t,n} = G_n\left(\frac{t}{n}, \varepsilon_t\right) \in \mathbb{R}^d, \quad t = 1, \dots, n \quad (1)$$

where $\varepsilon_t = (\epsilon_t, \epsilon_{t-1}, \dots) \in \mathbb{R}^\infty$ for iid random variables ϵ_i . For each $u \in [0, 1]$ and $n \in \mathbb{N}$, the sequence $(G_n(u, \varepsilon_t))_{t \in \mathbb{Z}}$ is stationary. By letting the kernel depend on the fraction $\frac{t}{n}$, the model explicitly accounts for nonstationarity.

The functions G_n are assumed to be measurable, where we endow the sequence space $\mathbb{R}^\infty$ with the projection σ -Algebra.

Assumption 2.1 (Limiting Kernel Convergence). *We assume that the kernel G_n tends towards a limiting kernel $G : \mathbb{R} \times \mathbb{R}^\infty \rightarrow \mathbb{R}^d$, in the sense that*

$$\sup_{u \in [0,1]} \|G_n(u, \varepsilon_0) - G(u, \varepsilon_0)\|_{L_q} \rightarrow 0, \quad n \rightarrow \infty$$

for some $q > 4$. Note that we do not require a rate of convergence in (2.1).

Assumption 2.2 (Lipschitz nonstationarity). *We require the function $u \mapsto G_n(u, \varepsilon_0) \in L_q(P)$ to satisfy specific regularity conditions. There exists a constant $C > 0$ such that for all $u, v \in [0, 1]$ and all $n \in \mathbb{N}$:*

$$\sup_n \|G_n(u, \varepsilon_0) - G_n(v, \varepsilon_0)\|_{L_q} \leq C|u - v|$$

Assumption 2.3 (Bounded dependence measure). *To define the dependence measure, we introduce an iid copy ϵ_i^* of the ϵ_i . Denote for $t \in \mathbb{Z}, j \in \mathbb{Z}_{\geq 0}$,*

$$\begin{aligned} \tilde{\varepsilon}_{t,j} &= (\epsilon_t, \dots, \epsilon_{t-j+1}, \epsilon_{t-j}^*, \epsilon_{t-j-1}, \dots) \in \mathbb{R}^\infty, \\ \varepsilon_{t,j}^* &= (\epsilon_t, \dots, \epsilon_{t-j+1}, \epsilon_{t-j}^*, \epsilon_{t-j-1}^*, \dots) \in \mathbb{R}^\infty. \end{aligned}$$

We assume that there exists a value $\rho \in (0, 1)$ such that, for all $j, n \in \mathbb{N}, u \in [0, 1]$,

$$\|G_n(u, \varepsilon_t) - G_n(u, \tilde{\varepsilon}_{t,j})\|_{L_q} \leq C_G \rho^j. \quad (2)$$

Assumption 2.4 (Identifiability). *The function $H : \mathbb{R}^d \times \mathbb{R}^p \rightarrow \mathbb{R}^p$, the parameter space $\Theta \subset \mathbb{R}^p$, and the class of distributions $\mathcal{P}$ satisfy: For any random vector $X \sim P \in \mathcal{P}$, there exists exactly one $\theta(P) \in \Theta$ such that $\mathbb{E}H(X, \theta(P)) = 0$.*

With this definition, we can speak of the local parameter $\theta_{t,n}$ as the solution of the moment equation $\mathbb{E}H(X_{t,n}, \theta_{t,n}) = 0$, that is, $\theta_{t,n} = \theta(P_{t,n}^X)$, and θ_u as the solution of $\mathbb{E}H(G(u, \varepsilon_0), \theta) = 0$.

This formulation in terms of nonlinear moment equations is very general and encompasses various examples of interest, as shown below.

Example 2.1 (Variance). *First, we consider the variance $\sigma^2 = \text{Var}(X)$. In this case, we also need the first moment $\mu = \mathbb{E}(X)$. We define the parameter vector $\theta \in \mathbb{R}^2$ as*

$$\theta = \begin{pmatrix} \mu \\ \sigma^2 \end{pmatrix} \leftrightarrow \begin{pmatrix} \mathbb{E}(X) \\ \text{Var}(X) \end{pmatrix}.$$

Thus, θ is the solution of the estimating equation $\mathbb{E}H(X, \theta) = 0$, where

$$H(X, \theta) = \begin{pmatrix} X - \mu \\ X^2 - \mu^2 - \sigma^2 \end{pmatrix}.$$

Example 2.2 (Autocorrelation). *For a time series Z_t not necessarily stationary but with $\mathbb{E}(Z_t^2) < \infty$ for all t , define $X_t = (Z_t, Z_{t-h})$, and consider the parameter vector $\theta \in \mathbb{R}^6$ and the corresponding estimating function $H(X, \theta)$ as*

$$\theta = \begin{pmatrix} \mu_1 \\ \mu_2 \\ \mu_{1,2} \\ \sigma_1^2 \\ \sigma_2^2 \\ \rho_h \end{pmatrix} \leftrightarrow \begin{pmatrix} \mathbb{E}(Z_t) \\ \mathbb{E}(Z_{t-h}) \\ \mathbb{E}(Z_t Z_{t-h}) \\ \text{Var}(Z_t) \\ \text{Var}(Z_{t-h}) \\ \text{Corr}(Z_t, Z_{t-h}) \end{pmatrix}, \quad H(X_t, \theta) = \begin{pmatrix} Z_t - \mu_1 \\ Z_{t-h} - \mu_2 \\ Z_t Z_{t-h} - \mu_{1,2} \\ Z_t^2 - \mu_1^2 - \sigma_1^2 \\ Z_{t-h}^2 - \mu_2^2 - \sigma_2^2 \\ \mu_{1,2} - \mu_1 \mu_2 - \rho_h \sigma_1 \sigma_2 \end{pmatrix}.$$

Then the autocovariance $\gamma(t, t-h) = \text{Cov}(Z_t, Z_{t-h})$ may be identified as $\gamma(t, t-h) = \theta(X_t)$.

Example 2.3 (Smoothed EDC density). *The smoothed extremal downside correlation (SEDC) matrix of a d -variate random vector X is defined as*

$$\text{SEDC}_{\tau, ij} = \text{Cor}(Y_i, Y_j),$$

where $Y_i = \frac{X_i - \mu_i}{\sigma_i} S_\gamma \left(\tau - \frac{X_i - \mu_i}{\sigma_i} \right), \quad \mu_i = \mathbb{E}(X_i), \quad \sigma_i^2 = \text{Var}(X_i).$

Here, $S_\gamma(\cdot)$ is the continuously differentiable logistic function $S_\gamma(x) = 1/(1 + \exp(-x/\gamma))$. The parameter $\gamma > 0$ is a smoothing parameter, and $\gamma \rightarrow 0$, $S_\gamma(x)$ converges to the Heaviside function. Based on the SEDC matrix, we define the SEDC density as

$$\text{SEDCD}_\tau = \frac{1}{d(d-1)} \sum_{i \neq j} |\text{SEDC}_{\tau, ij}|.$$

This statistical functional can be formulated in terms of estimating equations by defining the parameter vector $\theta \in \mathbb{R}^{d^2+3d+1}$ as

$$\theta = \begin{pmatrix} \text{SEDCD} \\ (\mu_i)_{i=1, \dots, d} \\ (\sigma_i^2)_{i=1, \dots, d} \\ (\nu_i)_{i=1, \dots, d} \\ (\gamma_{ij})_{i,j=1, \dots, d} \end{pmatrix} \leftrightarrow \begin{pmatrix} \text{SEDCD}_\tau \\ \mathbb{E}(X_i)_{i=1, \dots, d} \\ \text{Var}(X_i)_{i=1, \dots, d} \\ \mathbb{E}(Y_i)_{i=1, \dots, d} \\ \text{Cov}(Y_i, Y_j)_{i,j=1, \dots, d} \end{pmatrix}.$$

Accordingly, we define the estimating function H as

$$H(X, \theta) = \begin{pmatrix} (X_i - \mu_i)_{i=1, \dots, d} \\ (X_i^2 - \mu_i^2 - \sigma_i^2)_{i=1, \dots, d} \\ (Y_i - \nu_i)_{i=1, \dots, n} \\ (Y_i Y_j - \nu_i \nu_j - \gamma_{ij})_{i,j=1, \dots, d} \\ \text{SEDCD} - \frac{1}{d(d-1)} \sum_{i \neq j} \left| \frac{\gamma_{i,j}}{\gamma_{0,0}} \right| \end{pmatrix}.$$

The moment equation $\mathbb{E}H(X, \theta)$ may be solved explicitly by forward substitution, and we have $\theta_1 = \text{SED}CD_\tau$. Moreover, based on observations $X_1, \dots, X_n$, the empirical estimating equation $\frac{1}{n} \sum_{t=1}^n H(X_t, \hat{\theta})$ can be solved by substituting all moments in the SEDCD definition by empirical moments, which is also straightforward. Thus, the formulation in terms of estimating equations yields the same result as a naive estimator. However, the benefit of the implicit formulation is that it streamlines mathematical treatment and allows for a unified asymptotic theory, covering other examples beyond the extremal downside correlation.

Example 2.4 (Robust location). Consider a sample $X_1, X_2, \dots, X_n \in \mathbb{R}$ from distribution P , then an M-estimator for location can be found as a solution of the following problem:

$$\mu_t = \min_{\mu} \mathbb{E} \rho(X_t - \mu)$$

where $\rho : \mathbb{R} \rightarrow [0, \infty)$. For example, with $\rho(x) = x^2$, we obtain the mean, and with $\rho(x) = |x|$, we obtain the median. If ρ is differentiable, then it is possible to derive the location using the empirical estimating equation as follows:

$$\min_{\mu} \sum_{t=1}^n \rho(X_t - \mu) \implies \sum_{t=1}^n \psi(X_t - \mu) = 0$$

where $\psi(u) = \frac{d\rho(u)}{du}$ and $H(X, \mu) = \psi(X - \mu)$. The advantage of using this setting is that it is possible to define a function ρ that lies in the middle of the two cases discussed so far.

Example 2.5 (Time-Varying M-Regression). Let us consider a non-stationary time series $D_n = \{(X_t, Y_t)\}_{t=1}^n$, where $X_t \in \mathbb{R}^d$ denotes the vector of predictors and $Y_t \in \mathbb{R}$ is a univariate response series. We model the dynamic linear relationship via time-varying regression coefficients $\beta_{t,n} = \beta(\frac{t}{n}) \in \mathbb{R}^d$ subject to a local scale $\sigma_{t,n} = \sigma(\frac{t}{n}) > 0$.

The local parameter $\beta_{t,n}$ is identified as the root of the population estimating equation:

$$\mathbb{E}[H(D_t, \beta_{t,n})] = \mathbb{E} \left[X_t \psi \left(\frac{Y_t - X_t^\top \beta_{t,n}}{\sigma_{t,n}} \right) \right] = 0,$$

where $H : (\mathbb{R}^d \times \mathbb{R}) \times \mathbb{R}^d \rightarrow \mathbb{R}^d$ is the vector estimating function. In practice, $\sigma_{t,n}$ can be estimated through a localized robust pilot estimator, such as the normalized median absolute deviation (MAD) of preliminary residuals or estimated simultaneously by augmenting the system with a dedicated scale estimating equation. The score function $\psi(\cdot)$ can be chosen as either monotone or redescending. A key advantage of redescending score functions is that they completely suppress the influence of severe outliers, unlike monotone score functions, where the influence is still persistent. A caveat of redescending score functions is that they can lead to multiple local minima, making it challenging the identification of the optimal parameter. However, a consistent estimate of the roots can be computed using optimization procedure such as the Iteratively Reweighted Least Squares (IRWLS), which should be initialized with a robust pilot estimator for $\beta_{t,n}$ (such as an L_1 fit or an S-estimator).

For the asymptotic theory developed in the sequel, we need to impose certain regularity assumptions on the estimating function H .

Assumption 2.5. The parameter set Θ is open, and the function H satisfies the following properties:

- (i) The function $\theta \mapsto H(x, \theta)$ is three times differentiable such that $\sup_{P \in \mathcal{P}, \theta \in \Theta} \mathbb{E}_P \|D_\theta^k H(X, \theta)\|^q < \infty$ for $q = ???$ and $k = 0, 1, 2, 3$.
- (ii) The function H is locally Lipschitz in x such that $\sup_{\theta \in \Theta} \|D_\theta^k H(x, \theta) - D_\theta^k H(u, \theta)\| \leq [L(\|x\|) + L(\|y\|)] \cdot \|x - u\|$ for $k = 0, 1$, and the Lipschitz bound satisfies $\sup_{P \in \mathcal{P}} \mathbb{E}_P L(X)^4 < \infty$.
- (iii) The matrices $\mathbb{E}_P D_\theta H(X, \theta(P)) \in \mathbb{R}^{p \times p}$ are uniformly regular such that $\sup_{P \in \mathcal{P}} \|[\mathbb{E}_P D_\theta H(X, \theta(P))]^{-1}\| < \infty$.

Proposition 1. Under Assumptions 2.1, 2.2, 2.3, 2.4, 2.5, there exists a $C > 0$ such that

$$\|\theta_{t,n} - \theta_{r,n}\| \leq C \frac{|t-r|}{n}, \quad \|\theta_u - \theta_v\| \leq C|u - v|, \quad \sup_{u \in [0,1]} \|\theta_u - \theta_{\lfloor un \rfloor, n}\| \rightarrow 0.$$

Proof of Proposition 1. Let $X \sim P$ and $Y \sim Q$ be two arbitrary coupled random vectors. Then, for some $\tilde{\theta}$ between $\theta(P)$ and $\theta(Q)$,

$$0 = \mathbb{E}H(X, \theta(P)) - \mathbb{E}H(Y, \theta(Q)) = \mathbb{E}H(X, \theta(Q)) - \mathbb{E}H(Y, \theta(Q)) + [\mathbb{E}DH(X, \tilde{\theta})][\theta(P) - \theta(Q)].$$

By assumption (2.5), in particular the regularity condition (iii),

$$\begin{aligned} \|\theta(P) - \theta(Q)\| &\leq C\mathbb{E}\{[L(\|X\|) + L(\|Y\|)]\|X - Y\|\} \\ &\leq C\|X - Y\|_{L_2}. \end{aligned}$$

Now, the first inequality of the Proposition is obtained with $X = G_n(\frac{t}{n}, \varepsilon_0)$ and $Y = G_n(\frac{r}{n}, \varepsilon_0)$, such that $\|X - Y\|_{L_2} \leq C\frac{|t-r|}{n}$ by Assumption 2.1. The other bounds are derived analogously. $\square$

In our framework, we adopt the Dennis–Welsch score function (Holland and Welsch, 1977), defined as

$$\psi(u) = u \exp\left(-\left(\frac{u}{c}\right)^2\right),$$

where $c > 0$ is a tuning constant calibrated to achieve a target asymptotic Gaussian efficiency (e.g., $c = 2.985$ yields 95% efficiency).

Crucially, the Welsch score $\psi(u)$ is infinitely differentiable (C^∞) on the entire real line $\mathbb{R}$. Consequently, the estimating function $\theta \mapsto H(D_t, \theta)$ admits continuous derivatives of all orders, strictly fulfilling the C^3 differentiability and moment bounds required by Assumption 2.5(i).

3 Estimation

Recall that the parameter $\theta_{t,n}$ solves the (local) moment equation $\mathbb{E}H(X_{t,n}, \theta_{t,n}) = 0$. This local parameter can be estimated by $\hat{\theta}_{t,n}$ which solves the local estimating equation $H(\hat{\theta}_{t,n}) = 0$, where

$$H_{t,n}(\theta) = \frac{1}{M} \sum_{i=0}^{M-1} H(X_{t-i,n}, \theta), \quad (3)$$

and M is a bandwidth parameter.

Proposition 2 (Consistency of local estimates). *Suppose Assumptions 2.1, 2.2, 2.3, 2.4, 2.5 hold, for some $q > p$. Suppose furthermore that $M = M_n \rightarrow \infty$ such that*

$$n^{\frac{1}{q}} \left(\frac{1}{\sqrt{M_n}} + \frac{M_n}{n} \right) \rightarrow 0. \quad (4)$$

Then there exists a sequence of events $\mathcal{A}_n$ with $P(\mathcal{A}_n) \rightarrow 1$ such that the following hold:

- (i) *For any t there is a unique solution $\hat{\theta}_{t,n}$ of the local estimating equation.*
- (ii) *For any $\delta_n \rightarrow \infty$ arbitrarily slowly, there exists a sufficiently large C such that the maximum estimation error is bounded as*

$$\sup_t \|\hat{\theta}_{t,n} - \theta_{t,n}\| \leq C\delta_n n^{\frac{1}{q}} \left(\frac{1}{\sqrt{M}} + \frac{M}{n} \right).$$

- (iii) *The average estimation error is bounded as $\frac{1}{n} \sum_{t=1}^n \|\hat{\theta}_{t,n} - \theta_{t,n}\| \cdot \mathbf{1}_{\mathcal{A}_n}\|_{L_2}^2 \leq C(\frac{1}{M} + \frac{M^2}{n^2})$.*

- (iv) *There exists C such that $\sup_t \sup_{\theta \in \Theta} \|D_\theta^2 H_{t,n}\| \leq C$.*

- (v) *There exists C such that $\sup_t \|[DH_{t,n}(\hat{\theta}_{t,n})]^{-1}\| \leq C$.*

Remark 3.1. *The result of Proposition 2 does not ensure existence of $\hat{\theta}_{t,n}$ in all cases, but with increasing probability. That is, it might occur in finite samples that a local estimating equation does not admit a solution. Whether this scenario really arises depends also on the estimating function H . For example, $H(x, \theta) = x - \theta$ always admits a solution. The same holds for the variance in Example 2.1, for the robust location and for the robust regression in Examples 2.4 and 2.5. On the other hand, the autocorrelation in Example 2.2 and the SEDCD in Example 2.3 fail to exist if the variance in the denominator is zero, and this edge case needs separate handling. Thus, the generic assumptions on H formulated here can not ensure global existence.*

Proof of Proposition 2. Step (i): For any θ , define the time series $Y_{t,n}(\theta) = H(X_{t,n}, \theta) - \mathbb{E}H(X_{t,n}, \theta)$. Since H is Lipschitz, the time series $Y_{t,n}$ admits the same dependence bound as $X_{t,n}$, see Assumption 2.3. Thus, a Rosenthal inequality applies ([?, 3.2]), which states that the L_q norm of a sum scales as $\|\sum_{t=k+1}^{k+m} Y_{t,n}(\theta_{t,n})\|_{L_p} \leq C\sqrt{m}$, for any k and m . Thus,

$$\begin{aligned} H_{t,n}(\theta_{t,n}) &= \frac{1}{M} \sum_{i=0}^{M-1} H(X_{t-i,n}, \theta_{t,n}) \\ &= \frac{1}{M} \sum_{i=0}^{M-1} [H(X_{t-i,n}, \theta_{t,n}) - \mathbb{E}H(X_{t-i,n}, \theta_{t,n})] \\ &\quad + \frac{1}{M} \sum_{i=0}^{M-1} \left[\mathbb{E}H(X_{t-i,n}, \theta_{t,n}) - \underbrace{\mathbb{E}H(X_{t-i,n}, \theta_{t-i,n})}_{=0} \right] \\ &= \mathcal{O}_{L_q} \left(\frac{1}{\sqrt{M}} + \frac{M}{n} \right), \end{aligned}$$

upon using Proposition 1 and the regularity of H in the last step. Using the higher-order regularity of H , the same arguments show that, for any $\theta, \theta' \in \Theta$

$$\|D_\theta H_{t,n}(\theta) - \mathbb{E}D_\theta H(X_{t,n}, \theta)\|_{L_q} \leq C \cdot \left(\frac{1}{\sqrt{M}} + \frac{M}{n} \right),$$

and, coordinate-wise for $j, k = 1, \dots, p$,

$$\begin{aligned} &\left\| [D_\theta H_{t,n}(\theta) - \mathbb{E}D_\theta H(X_{t,n}, \theta)]_{jk} - [D_\theta H_{t,n}(\theta') - \mathbb{E}D_\theta H(X_{t,n}, \theta')]_{jk} \right\|_{L_q} \\ &= \|D_\theta [D_\theta H_{t,n}(\tilde{\theta}_{jk}) - \mathbb{E}D_\theta H_{t,n}(\tilde{\theta}_{jk})]_{jk} \cdot (\theta - \theta')\|_{L_q} \leq C \|\theta - \theta'\| \left(\frac{1}{\sqrt{M}} + \frac{M}{n} \right). \end{aligned}$$

This is sufficient to apply an L_q -chaining argument to the process $L(\theta) = \|D_\theta H_{t,n}(\theta) - \mathbb{E}D_\theta H_{t,n}(\theta)\|$, and metric $d(\theta, \theta') = \|L(\theta) - L(\theta')\|_{L_q}$. Then any bounded subset $\Theta^* \subset \Theta$ has covering numbers of the form

$$N(\Theta^*, \epsilon, d) \leq C \left[\left(\frac{1}{\sqrt{M}} + \frac{M}{n} \right) \frac{\text{diam}(\Theta^*)}{\epsilon} \right]^p$$

and thus L_q chaining yields

$$\begin{aligned} \left\| \sup_{\theta \in \Theta^*} \|D_\theta H_{t,n}(\theta) - \mathbb{E}D_\theta H(X_{t,n}, \theta)\| \right\|_{L_q} &= \left\| \sup_{\theta \in \Theta^*} L(\theta) \right\|_{L_q} \leq C \int_0^{C \left(\frac{1}{\sqrt{M}} + \frac{M}{n} \right) \text{diam}(\Theta^*)} N(\Theta^*, \epsilon, d)^{1/q} d\epsilon \\ &= C \left(\frac{1}{\sqrt{M}} + \frac{M}{n} \right) \text{diam}(\Theta^*) \int_0^1 \epsilon^{-\frac{p}{q}} d\epsilon, \end{aligned}$$

which is finite for $q > p$. With an analogous chaining argument, we can show that

$$\left\| \sup_{\theta \in \Theta^*} \|H_{t,n}(\theta) - \mathbb{E}H_{t,n}(\theta)\| \right\|_{L_q} \leq C \left(\frac{1}{\sqrt{M}} + \frac{M}{n} \right) \text{diam}(\Theta^*).$$

and the same holds for the second derivatives. We conclude that

$$\begin{aligned} \left\| \sup_t \sup_{\theta \in \Theta^*} \|H_{t,n}(\theta) - \mathbb{E}H_{t,n}(\theta)\| \right\|_{L_q} &\leq C n^{\frac{1}{q}} \left(\frac{1}{\sqrt{M}} + \frac{M}{n} \right) =: \Delta_n, \\ \left\| \sup_t \sup_{\theta \in \Theta^*} \|D_\theta H_{t,n}(\theta) - \mathbb{E}D_\theta H(X_{t,n}, \theta)\| \right\|_{L_q} &\leq \Delta_n, \\ \left\| \sup_t \sup_{\theta \in \Theta^*} \|D_\theta^2 H_{t,n}(\theta) - \mathbb{E}D_\theta^2 H(X_{t,n}, \theta)\| \right\|_{L_q} &\leq \Delta_n. \end{aligned}$$

For some δ_n , define the event

$$\begin{aligned} \mathcal{A}_n = \Big\{ &\sup_t \sup_{\theta \in \Theta^*} \|H_{t,n}(\theta) - \mathbb{E}H_{t,n}(\theta)\| \leq \delta_n \\ &\text{and } \sup_t \sup_{\theta \in \Theta^*} \|D_\theta H_{t,n}(\theta) - \mathbb{E}D_\theta H(X_{t,n}, \theta)\| \leq \delta_n \\ &\text{and } \sup_t \sup_{\theta \in \Theta^*} \|D_\theta^2 H_{t,n}(\theta) - \mathbb{E}D_\theta^2 H(X_{t,n}, \theta)\| \leq \delta_n \Big\}. \end{aligned}$$

If $1 \gg \delta_n \gg \Delta_n$, we have $P(\mathcal{A}_n) \rightarrow 1$. Since $\Delta_n \rightarrow 0$ by assumption (4), this choice of δ_n is possible.

Step (ii): Equivalently to solving $H_{t,n}(\theta) = 0$, we may look for a fixed point of the function $\phi_{t,n}(\theta) = \theta - [D_\theta H_{t,n}(\theta_{t,n})]^{-1} H_{t,n}(\theta)$, and we proceed to show that this mapping is a contraction on the ball $B_{r_n}(\theta_{t,n})$, with rate r_n determined later. Banach's Theorem then yields the existence of a fixed point. This would imply that (i) there is a solution $H_{t,n}(\hat{\theta}_{t,n}) = 0$, and (ii) this solution satisfies $\|\hat{\theta}_{t,n} - \theta_{t,n}\| \leq r_n$.

To establish that $\phi_{t,n}$ is a contraction, we compute

$$\begin{aligned} D_\theta \phi_{t,n}(\theta) &= I - [D_\theta H_{t,n}(\theta_{t,n})]^{-1} D_\theta H_{t,n}(\theta) \\ &= I - [D_\theta H_{t,n}(\theta_{t,n})]^{-1} [D_\theta H_{t,n}(\theta) - D_\theta H_{t,n}(\theta_{t,n}) + D_\theta H_{t,n}(\theta_{t,n})] \\ &= [D_\theta H_{t,n}(\theta_{t,n})]^{-1} [D_\theta H_{t,n}(\theta) - D_\theta H_{t,n}(\theta_{t,n})]. \end{aligned}$$

Choose r_n such that

$$\sup_{\theta \in B_{r_n}(\theta_{t,n})} \left\| [D_\theta H_{t,n}(\theta_{t,n})]^{-1} \underbrace{[D_\theta H_{t,n}(\theta) - D_\theta H_{t,n}(\theta_{t,n})]}_{=\mathcal{O}(r_n)} \right\| \leq \frac{1}{4}.$$

Note that the inverse is bounded by Assumption 2.5. Hence, there exists some $r_0 > 0$ such that the previous assertion holds for all $r_n \leq r_0$. If n is big enough such that δ_n is small enough, we may conclude that on the event $\mathcal{A}_n$, it holds $\sup_{\theta \in B_{r_n}(\theta_{t,n})} \|D_\theta \phi_{t,n}(\theta)\| \leq \frac{1}{2}$. Thus, $\phi_{t,n}$ is a contraction. Then define the recursion $\theta^{(0)} = \theta_{t,n}$, and $\theta^{(k+1)} = \phi_{t,n}(\theta^{(k)})$, such that

$$\begin{aligned} \|\theta^{(k+1)} - \theta^{(0)}\| &\leq \sum_{r=1}^k \|\theta^{(r+1)} - \theta^{(r)}\| \leq \sum_{r=1}^\infty 2^{-r} \|\phi_{t,n}(\theta_{t,n}) - \theta_{t,n}\| \\ &= \|[D_\theta H_{t,n}(\theta_{t,n})]^{-1} H_{t,n}(\theta_{t,n})\|. \end{aligned}$$

On the event $\mathcal{A}_n$, for n large enough such that δ_n is small enough, the inverse is upper bounded, and $H_{t,n}(\theta_{t,n}) = \mathbb{E}H_{t,n}(\theta_{t,n}) + \mathcal{O}(\delta_n) = 0 + \mathcal{O}(\delta_n) + \mathcal{O}(M/n)$. Now choose r_n such that $1 \gg r_n \geq C(\delta_n + \frac{M}{n})$. Then, $\|\theta^{(k+1)} - \theta^{(0)}\| \leq C\delta_n + C\frac{M}{n} \leq r_n$ for n large enough. Hence, we conclude that $\theta^{(k)} \in B_{r_n}(\theta_{t,n})$ for all k , and thus the contraction yields that $\theta^{(k)} \rightarrow \theta^*$ as $k \rightarrow \infty$, and the limit solves $\phi_{t,n}(\theta^*) = \theta^*$. This is the solution of the estimating equation, i.e. $\hat{\theta}_{t,n} = \theta^*$, satisfying $\|\hat{\theta}_{t,n} - \theta_{t,n}\| \leq r_n$. Banach's Theorem, with r_0 instead of r_n , also yields uniqueness on $B_{r_0}(\theta_{t,n})$. We have thus established the existence of $\hat{\theta}_{t,n}$ and its uniform rate of convergence.

Step (iii): On the event $\mathcal{A}_n$, we have, as shown above, $\|\hat{\theta}_{t,n} - \theta_{t,n}\| \leq C\|H_{t,n}(\theta_{t,n})\|$, which is bounded in L_q by $\frac{1}{\sqrt{M}} + \frac{M}{n}$. Now, outside of the event $\mathcal{A}_n$ where $\hat{\theta}_{t,n}$ is well-defined, we set $\hat{\theta}_{t,n} = \theta_{t,n}$. With this convention, we may conclude that

$$\frac{1}{n} \sum_{t=1}^n \|\hat{\theta}_{t,n} - \theta_{t,n}\|_{L_2}^2 = \mathcal{O}\left(\frac{1}{M} + \frac{M^2}{n^2}\right).$$

Step (iv): Claims (iv) is an immediate consequence of the bound on $D_\theta^2(X_{t,n}, \theta_{t,n})$ and the uniform convergence on the event $\mathcal{A}_n$. Claim (v) follows also via the uniform convergence, and the uniform regularity of $\mathbb{E}D_\theta H(X_{t,n}, \theta_{t,n})$. $\square$

Consider the linearized partial sum estimator

$$\hat{\Theta}_n^{\text{LIN}}(u) = \frac{1}{n} \sum_{t=1}^{\lfloor un \rfloor} \hat{\theta}_{t,n} - DH_{t,n}(\hat{\theta}_{t,n})^{-1} H(X_{t+L}, \hat{\theta}_{t,n}),$$

where D denotes the Jacobian with respect to θ , and L is an offset to be specified later. Moreover, we define $\Theta_n(u) = \frac{1}{n} \sum_{t=1}^{\lfloor un \rfloor} \theta_{t+L,n}$.

TODO: Fix the initial offset, in the definition and the proofs

TODO: Relax Lipschitz assumption to Hölder regularity, to illustrate the benefit of linearization

Theorem 3.1 (Functional CLT). *Suppose Assumptions 2.1, 2.2, 2.3, 2.4, 2.5 hold for some $q > \max(3, p)$, and $L = L_n$ and $M = M_n$ are chosen such that*

$$\log(n) \ll L_n \ll M_n \ll n^{\frac{2}{3}}, \quad M_n \gg n^{\frac{2}{q}}, \quad L_n \ll n^{\frac{1}{2} - \frac{1}{q}}. \quad (5)$$

Then, weakly in the Skorokhod space $D([0, 1], \mathbb{R}^p)$,

$$\sqrt{n} \left[\widehat{\Theta}_n^{\text{LIN}}(u) - \Theta_n(u) \right] \Rightarrow \int_0^u g_v \Sigma_v^{\frac{1}{2}}, dW_v,$$

where W_v is a standard Brownian motion in $\mathbb{R}^p$, and

$$g_u = [\mathbb{E}DH(G(u, \varepsilon_0), \theta_u)]^{-1}, \quad \Sigma_u = \sum_{h=-\infty}^{\infty} \text{Cov}(G(u, \varepsilon_0), G(u, \varepsilon_h)).$$

Proof of Theorem 3.1. The pilot $\widehat{\theta}_{t,n}$ satisfies $H_{t,n}(\widehat{\theta}_{t,n}) = 0$ by definition (3), and thus

$$\begin{aligned} H_{t,n}(\theta_{t+L,n}) &= H_{t,n}(\theta_{t+L,n}) - H_{t,n}(\widehat{\theta}_{t,n}) \\ &= -DH_{t,n}(\widehat{\theta}_{t,n}) \left[\widehat{\theta}_{t,n} - \theta_{t+L,n} \right] + \|D^2 H_{t,n}(\widetilde{\theta}_t)\| \|\theta_{t+L,n} - \widehat{\theta}_{t,n}\|^2, \end{aligned}$$

for some $\widetilde{\theta}_t$ on the line between $\theta_{t+L,n}$ and $\widehat{\theta}_{t,n}$, by the mean value theorem. Introduce the event

$$\mathcal{A}_n = \bigcap_{t=1}^n \left\{ \| [DH_{t,n}(\widehat{\theta}_{t,n})]^{-1} \| \leq C \quad \text{and} \quad \sup_{\theta \in \Theta} \|D^2 H_{t,n}(\theta)\| \leq C \quad \text{and} \quad \widehat{\theta}_{t,n} \in \Theta \right\}.$$

Proposition 2 shows that $P(\mathcal{A}_n) \rightarrow 1$ as $n \rightarrow \infty$ for C large enough, as a consequence of the rate constraints (5). Thus, in the sequel, we may always suppose that the guarantees of $\mathcal{A}_n$ hold, and define $\widehat{\theta}_{t,n} = \theta_{t,n}$ outside of $\mathcal{A}_n$ without altering the asymptotics. In the case that $\mathcal{A}_n$ holds,

$$\widehat{\theta}_{t,n} - \theta_{t+L,n} = -DH_{t,n}(\widehat{\theta}_{t,n})^{-1} H_{t,n}(\theta_{t+L,n}) + \mathcal{O}(\|\theta_{t+L,n} - \widehat{\theta}_{t,n}\|^2),$$

and thus

$$\begin{aligned} &\widehat{\Theta}_n(u) - \Theta_n(u) \\ &= \frac{1}{n} \sum_{t=1}^{\lfloor un \rfloor} -DH_{t,n}(\widehat{\theta}_{t,n})^{-1} H_{t,n}(\theta_{t+L,n}) - DH_{t,n}(\widehat{\theta}_{t,n})^{-1} H(X_{t+L}, \widehat{\theta}_{t,n}) \\ &\quad + \mathcal{O}\left(\frac{1}{n} \sum_{t=1}^n \|\widehat{\theta}_{t,n} - \theta_{t+L,n}\|^2\right) \\ &= -\frac{1}{n} \sum_{t=1}^{\lfloor un \rfloor} DH_{t,n}(\widehat{\theta}_{t,n})^{-1} \left[H(X_{t+L}, \widehat{\theta}_{t,n}) + H_{t,n}(\theta_{t+L,n}) \right] + R_n \end{aligned}$$

where R_n collects all remainder terms to be studied later,

$$\begin{aligned} &= -\frac{1}{n} \sum_{t=1}^{\lfloor un \rfloor} DH_{t,n}(\widehat{\theta}_{t,n})^{-1} H(X_{t+L}, \theta_{t+L,n}) \\ &\quad - \underbrace{\frac{1}{n} \sum_{t=1}^{\lfloor un \rfloor} DH_{t,n}(\widehat{\theta}_{t,n})^{-1} \left[H(X_{t+L}, \widehat{\theta}_{t,n}) - H(X_{t+L}, \theta_{t+L,n}) + H_{t,n}(\theta_{t+L,n}) - \overbrace{H_{t,n}(\widehat{\theta}_{t,n})}^{=0} \right]}_{=B_n} + R_n \\ &= \frac{1}{n} \sum_{t=1}^{\lfloor un \rfloor} DH_{t,n}(\widehat{\theta}_{t,n})^{-1} H(X_{t+L}, \theta_{t+L,n}) + B_n + R_n = A_n(u) + B_n(u) + R_n(u). \end{aligned} \tag{6}$$

We will show that $\sqrt{n}A_n$ converges to a Gaussian process, while B_n and R_n are negligible. In particular, the fact that $\|\theta_{t+L,n} - \theta_{t,n}\| \leq C \frac{L}{n}$, and Proposition 2(iii), yield the bound

$$R_n = \mathcal{O}_P\left(\frac{1}{M} + \frac{M^2 + L^2}{n^2}\right). \tag{7}$$

Regarding B_n , on the event $\mathcal{A}_n$, linearization yields

$$\begin{aligned}
B_n(u) &= \frac{1}{n} \sum_{t=1}^{\lfloor un \rfloor} DH_{t,n}(\hat{\theta}_{t,n})^{-1} [DH(X_{t+L}, \theta_{t+L,n}) - DH_{t,n}(\theta_{t+L,n})] (\hat{\theta}_{t,n} - \theta_{t+L,n}) \\
&\quad + \mathcal{O} \left(\frac{1}{n} \sum_{t=1}^n \|\hat{\theta}_{t,n} - \theta_{t+L,n}\|^2 \right) \\
&= \frac{1}{n} \sum_{t=1}^{\lfloor un \rfloor} DH_{t,n}(\hat{\theta}_{t,n})^{-1} \left[DH(X_{t+L}, \theta_{t+L,n}) - \frac{1}{M} \sum_{i=0}^{M-1} DH(X_{t-i}, \theta_{t+L,n}) \right] (\hat{\theta}_{t,n} - \theta_{t+L,n}) \\
&\quad + \mathcal{O} \left(\frac{1}{n} \sum_{t=1}^n \|\hat{\theta}_{t,n} - \theta_{t+L,n}\|^2 \right) \\
&= \frac{1}{n} \sum_{t=1}^{\lfloor un \rfloor} DH_{t,n}(\hat{\theta}_{t,n})^{-1} [DH(X_{t+L}, \theta_{t+L,n}) - E_{t+L,t+L}] (\hat{\theta}_{t,n} - \theta_{t+L,n}) \\
&\quad + \mathcal{O} \left(\sqrt{\frac{1}{n} \sum_{t=1}^n \|\hat{\theta}_{t,n} - \theta_{t+L,n}\|^2} \sqrt{\frac{1}{n} \sum_{t=1}^n \left\| \frac{1}{M} \sum_{i=0}^{M-1} DH(X_{t-i}, \theta_{t+L,n}) - E_{t+L,t+L} \right\|^2} \right) \\
&\quad + \mathcal{O} \left(\frac{1}{n} \sum_{t=1}^n \|\hat{\theta}_{t,n} - \theta_{t+L,n}\|^2 \right) \\
&= B_{n,1}(u) + B_{n,2} + R_n,
\end{aligned}$$

where $E_{s,t} = \mathbb{E}[DH(X_s, \theta_{t,n})]$. Assumption 2.2 and 2.5(i) imply $|E_{s,t} - E_{r,t}| \leq |r - s|/n$, which yields

$$\sqrt{\frac{1}{n} \sum_{t=1}^n \left\| \frac{1}{M} \sum_{i=0}^{M-1} E_{t-i,t+L} - E_{t+L,t+L} \right\|^2} \leq \frac{M+L}{n}.$$

Furthermore, by virtue of the local Lipschitz property in Assumption (2.2)(ii) and Hölder's inequality, we may bound the physical dependence measure of $DH(X, \theta)$ as

$$\begin{aligned}
&\sup_{\theta \in \Theta} \sup_{u \in [0,1]} \|DH(G_n(u, \varepsilon_0), \theta) - DH(G_n(u, \tilde{\varepsilon}_{0,j}), \theta)\|_{L_2} \\
&\leq \|L(G_n(u, \varepsilon_0))\|_{L_4} \|G_n(u, \varepsilon_0) - G_n(u, \tilde{\varepsilon}_{0,j})\|_{L_4} \\
&\leq C\rho^j,
\end{aligned}$$

using (2.3) in the last step. Hence, a concentration inequality exploiting the physical dependence measure, e.g. (? , Thm. 3.2), yields

$$\left\| \frac{1}{M} \sum_{i=0}^{M-1} DH(X_{t-i}, \theta_{t+L,n}) - E_{t-i,t+L} \right\|_{L_2} = \mathcal{O}(\frac{1}{\sqrt{M}}).$$

We thus found

$$B_{n,2} = \mathcal{O}_P(\sqrt{R_n}) \cdot \mathcal{O}_P\left(\frac{1}{\sqrt{M}} + \frac{M+L}{n}\right).$$

To handle $B_{n,1}$, introduce $X_{t+L}^* = G_n(\frac{t+L}{n}, \varepsilon_{t+L,n}^*)$, which is equal in distribution to X_{t+L} , but by construction independent of $\hat{\theta}_{t,n}$. Assumption (2.3) ensures that $\|X_{t+L} - X_{t+L}^*\|_{L_q} \leq C\rho^L$, and thus $\|DH(X_{t+L}, \theta_{t+L,n}) - DH(X_{t+L}^*, \theta_{t+L,n})\|_{L_2} \leq C\rho^L$. Hence,

$$B_{n,1}(u) = \frac{1}{n} \sum_{t=1}^{\lfloor un \rfloor} DH_{t,n}(\hat{\theta}_{t,n})^{-1} [DH(X_{t+L}^*, \theta_{t+L,n}) - E_{t+L,t+L}] (\hat{\theta}_{t,n} - \theta_{t+L,n}) + \mathcal{O}_P(\rho^L).$$

Now decompose the first term into blocks of length L as

$$\begin{aligned}
& \frac{1}{n} \sum_{t=1}^{\lfloor un \rfloor} DH_{t,n}(\hat{\theta}_{t,n})^{-1} [DH(X_{t+L}^*, \theta_{t+L,n}) - E_{t+L,t+L}] (\hat{\theta}_{t,n} - \theta_{t+L,n}) \\
&= \frac{1}{L} \sum_{k=1}^L \frac{L}{n} \sum_{\substack{t=iL+k \\ i=0, \dots, \lfloor n/L \rfloor}}^{\lfloor un \rfloor} DH_{t,n}(\hat{\theta}_{t,n})^{-1} [DH(X_{t+L}^*, \theta_{t+L,n}) - E_{t+L,t+L}] (\hat{\theta}_{t,n} - \theta_{t+L,n}).
\end{aligned} \tag{8}$$

Observe that each of the L terms is a martingale due the decoupling. We may thus bound

$$\begin{aligned}
& \left\| \sum_{\substack{t=iL+k \\ i=0, \dots, \lfloor n/L \rfloor}}^{\lfloor un \rfloor} DH_{t,n}(\hat{\theta}_{t,n})^{-1} [DH(X_{t+L}^*, \theta_{t+L,n}) - E_{t+L,t+L}] (\hat{\theta}_{t,n} - \theta_{t+L,n}) \right\|_{L_2}^2 \\
&= \sum_{\substack{t=iL+k \\ i=0, \dots, \lfloor n/L \rfloor}}^{\lfloor un \rfloor} \left\| DH_{t,n}(\hat{\theta}_{t,n})^{-1} [DH(X_{t+L}^*, \theta_{t+L,n}) - E_{t+L,t+L}] (\hat{\theta}_{t,n} - \theta_{t+L,n}) \right\|_{L_2}^2 \\
&\leq C \sum_{\substack{t=iL+k \\ i=0, \dots, \lfloor n/L \rfloor}}^{\lfloor un \rfloor} \left\| \hat{\theta}_{t,n} - \theta_{t+L,n} \right\|_{L_2}^2 \\
&\leq C \sum_{\substack{t=iL+k \\ i=0, \dots, \lfloor n/L \rfloor}}^{\lfloor un \rfloor} \left\| \hat{\theta}_{t,n} - \theta_{t,n} \right\|_{L_2}^2 + C \frac{n}{L} \left(\frac{L}{n} \right)^2
\end{aligned}$$

This bound holds uniformly in u due to Doob's maximal inequality. Returning to (8), this yields

$$\begin{aligned}
\sup_u |B_{n,1}(u)| &= \mathcal{O}_P \left(\frac{1}{n} \sum_{k=1}^L \sqrt{\sum_{\substack{t=iL+k \\ i=0, \dots, \lfloor n/L \rfloor}}^n \left\| \hat{\theta}_{t,n} - \theta_{t,n} \right\|_{L_2}^2 + \left(\frac{L}{n} \right)^{\frac{3}{2}} + \rho^L} \right) \\
&\leq \mathcal{O}_P \left(\sqrt{\frac{L}{n}} \sqrt{\frac{1}{n} \sum_{t=1}^n \left\| \hat{\theta}_{t,n} - \theta_{t,n} \right\|_{L_2}^2 + \left(\frac{L}{n} \right)^{\frac{3}{2}} + \rho^L} \right) \\
&= \mathcal{O}_P \left(\sqrt{\frac{L}{n}} \left(\frac{1}{\sqrt{M}} + \frac{M}{n} \right) + \left(\frac{L}{n} \right)^{\frac{3}{2}} + \rho^L \right),
\end{aligned}$$

using Proposition 2(iii) in the last step [check these rates again](#)

Overall, we found that

$$\begin{aligned}
\sup_u |B_n(u)| &= \mathcal{O}_P \left(\rho^L + \left(\frac{L}{n} \right)^{\frac{3}{2}} + \sqrt{\frac{L}{n}} \left(\frac{1}{\sqrt{M}} + \frac{M}{n} \right) + R_n + \sqrt{R_n} \left[\frac{1}{\sqrt{M}} + \frac{M+L}{n} \right] \right) \\
&= \mathcal{O}_P \left(\rho^L + \left(\frac{L}{n} \right)^{\frac{3}{2}} + \sqrt{\frac{L}{n}} \left(\frac{1}{\sqrt{M}} + \frac{M}{n} \right) + \left(\frac{M+L}{n} \right)^2 \right).
\end{aligned} \tag{9}$$

Obviously, if B_n is negligible, then so is R_n . Note that (9) is of order $o(1/\sqrt{n})$ if L, M are chosen such that, for some $\delta > 0$

$$\frac{\left(\frac{1}{2} + \delta \right)}{|\log \rho|} \log(n) \leq L_n \ll n^{\frac{2}{3}}, \quad L_n \ll M_n, \quad M_n \ll n^{\frac{3}{4}}, \quad L_n M_n^2 \ll n^2. \tag{10}$$

Moreover, to apply Proposition 2, we need to ensure that (4) tends to zero. This gives rise to the additional rate constraints

$$n^{\frac{2}{q}} \ll M_n \ll n^{1-\frac{1}{q}}. \tag{11}$$

The rate constraints (5) are simpler than (10) and (11), but stricter than the latter.

Returning to (6), we now establish weak convergence of $A_n(u)$. Notice that $\mathbb{E}H(X_{t+L}, \theta_{t+L,n}) = 0$ by definition. If the inverse matrix multiplier were non-random, we could thus apply a functional central limit theorem for partial sums of nonstationary time series. Fortunately, the inverse matrix concentrates around its expectation, which still allows to apply a functional CLT. This concept has been detailed in ?, and we specifically check the conditions of Theorem 5 therein. To this end, matching the notation therein, we define the random multipliers $\hat{g}_{t,n} = DH_{t,n}(\hat{\theta}_{t,n})^{-1}$, the pre-asymptotic multiplier $g_{t,n} = [\mathbb{E}DH(X_{t,n}, \theta_{t,n})]^{-1}$, and the asymptotic multiplier $g_u = [\mathbb{E}DH(G(u, \varepsilon_0), \theta(u))]^{-1}$. As the Jacobian is uniformly elliptic on the event $\mathcal{A}_n$, we find

$$\begin{aligned} \|g_u - g_{\lfloor un \rfloor, n}\| &\leq C \left\| \mathbb{E}DH(G(u, \varepsilon_0), \theta_u) - \mathbb{E}DH(G_n(\frac{\lfloor un \rfloor}{n}, \varepsilon_0), \theta_{\lfloor un \rfloor, n}) \right\| \\ &\leq C \left\| \mathbb{E}DH(G(u, \varepsilon_0), \theta_u) - \mathbb{E}DH(G(u, \varepsilon_0), \theta_{\lfloor un \rfloor, n}) \right\| \\ &\quad + C \left\| \mathbb{E}DH(G(u, \varepsilon_0), \theta_{\lfloor un \rfloor, n}) - \mathbb{E}DH(G_n(\frac{\lfloor un \rfloor}{n}, \varepsilon_0), \theta_{\lfloor un \rfloor, n}) \right\| \\ &\leq C \|\theta_u - \theta_{\lfloor un \rfloor, n}\| + C \|G(u, \varepsilon_0) - G_n(\frac{\lfloor un \rfloor}{n}, \varepsilon_0)\|_{L_2}. \end{aligned}$$

In the last step, we used the bounds on $D^2H(x, \theta)$ for the first term and the Lipschitz continuity of $x \mapsto DH(x, \theta)$ for the second term. Thus, $\sup_{u \in [0, 1]} \|g_u - g_{\lfloor un \rfloor, n}\| \rightarrow 0$, which implies condition (A.4) in (?). Note moreover that conditions (A.1), (A.2) and (A.3) therein are implied by our assumptions, and, using the notation therein, $\Psi_n, \Theta_n, \Gamma_n, \Phi_n = \mathcal{O}(1)$, and $\Xi_n = \mathcal{O}(\rho^L)$. Moreover, conditionally on the event $\mathcal{A}_n$,

$$\begin{aligned} \Lambda_n^2 &= \sum_{t=1}^n \|\hat{g}_{t,n} - g_{t,n}\|_{L_2}^2 \leq C \sum_{t=1}^n \|D_\theta H_{t,n}(\hat{\theta}_{t,n}) - \mathbb{E}D_\theta H(X_{t,n}, \theta_{t,n})\|_{L_2}^2 \\ &\leq C \sum_{t=1}^n \|D_\theta H_{t,n}(\theta_{t,n}) - \mathbb{E}D_\theta H(X_{t,n}, \theta_{t,n})\|_{L_2}^2 + C \sum_{t=1}^n \|\hat{\theta}_{t,n} - \theta_{t,n}\|_{L_2}^2 \end{aligned}$$

The first term can be controlled via the dependence bound in Assumption 2.3, showing that $\|D_\theta H_{t,n}(\theta_{t,n}) - \mathbb{E}D_\theta H(X_{t,n}, \theta_{t,n})\|_{L_2}^2 \leq C/M$. The second part may be controlled via Proposition 2(iii). Thus,

$$\Lambda_n^2 \leq C \frac{n}{M} + Cn \left(\frac{1}{M} + \frac{M^2}{n^2} \right) = \mathcal{O} \left(\frac{n}{M} + \frac{M^2}{n} \right).$$

To apply the multiplier FCLT (?, Thm. 5), we need to ensure that

$$\Lambda_n \left(\frac{1}{\sqrt{n}} + \Xi_n \right) \rightarrow 0, \quad L_n \ll n^{\frac{1}{2} - \frac{1}{q}}.$$

In view of the imposed lower bound on L_n , we have $\Xi_n \ll 1/\sqrt{n}$, such that $\Lambda_n \Xi_n \rightarrow 0$. Moreover, as $M_n \rightarrow \infty$ and $M^2 \ll n^{\frac{4}{3}}$, we have $\Lambda_n/\sqrt{n} \rightarrow 0$, validating all conditions. Applying the referenced FCLT, we thus find

$$\begin{aligned} \sqrt{n}A_n(u) &\Rightarrow \int_0^u g_v \Sigma_v^{\frac{1}{2}} dW_v = \int_0^u [\mathbb{E}DH(G(u, \varepsilon_0), \theta(u))]^{-1} \Sigma_v^{\frac{1}{2}} dW_v, \\ \text{where } \Sigma_v &= \sum_{h=-\infty}^{\infty} \text{Cov}[H(G(v, \varepsilon_0)), H(G(v, \varepsilon_h))]. \end{aligned}$$

The matrix Σ_v is the local long run covariance of the locally stationary time series $H(X_{t,n}, \theta_{t,n})$. $\square$

Theorem 3.2 (Bootstrap consistency). *Suppose the conditions of Theorem 3.1 hold for some moment $q > 4$. Choose a block sequence $b_n \rightarrow \infty$ such that $b_n = o(n^{1/3})$. Then, as $n \rightarrow \infty$,*

$$\begin{aligned} Q_n(u) &= \frac{1}{n} \sum_{t=\tau_n+L_n}^{\lfloor nu \rfloor - b_n} \frac{1}{b_n} \left[DH_{t,n}(\hat{\theta}_{t,n})^{-1} \sum_{i=L_n+1}^{L_n+b_n} H(X_{t+i,n}, \hat{\theta}_{t,n}) \right]^2 \\ &\xrightarrow{P} Q(u) = \int_0^u g_v \Sigma_v g_v^T dv. \end{aligned}$$

Proof of Theorem 3.2. **TODO** $\square$

Theorem 3.2 provides the theoretical justification for the Multiplier Bootstrap. Because $Q_n(u)$ uniformly converges to the true integrated variance $Q(u)$, we can use it to scale sequences of standard normal variables, generating artificial sample paths that mimic the asymptotic null distribution of the CUSUM test statistic.

4 CUSUM test

5 Finite sample performance

5.1 M-estimator of location

In this section, we showcase the advantages of the linearization step for detecting structural breaks in a robust location parameter. We design a Monte Carlo simulation study to evaluate the bias and Standard error of the proposed linearized estimator against a standard plug-in approach and test empirical size and power against alternative competitors for robust change in shift.

Let us consider a locally stationary Data Generating Process (DGP) for the time series $X_{t,n}$ over the fractional time index $u = \frac{t}{n}$ and $u_k = \frac{t-k}{n} \in [0, 1]$, defined as:

$$X_{t,n} = \mu(u) + \sum_{j=1}^p a_j X_{t-j,n} + \sigma(u)e_t + \sum_{k=1}^q b_k \sigma(u_k)e_{t-k}. \quad (12)$$

Here, a_j and b_k are strictly stationary AR(p) and MA(q) coefficients respectively. For this simulation, we consider an AR(2) process with $(a_1, a_2) = (0.2, -0.1)$ and an MA(1) process with $b_1 = 0.2$. The variable ϵ_t represents the white noise innovations, which we simulate using either standard Gaussian or heavy-tailed Student's t_3 distributions. To further evaluate the robustness of the proposed test, we introduce unconditional heteroskedasticity into the model. Although time-varying variance induces non-stationarity, a well-calibrated change-in-mean test must control the empirical size under the null hypothesis while retaining detection power under the alternative. Specifically, we allow the scale function $\sigma(u)$ to vary according to three scenarios:

- (i) **Smooth trending variance:** $\sigma(u) = \exp(u/2)$,
- (ii) **Smooth trending variance:** $\sigma(u) = \exp(u)$,
- (iii) **Cyclic variance:** $\sigma(u) = 1 + \sin(2\pi u)$,
- (iv) **Abrupt variance break:** $\sigma(u) = \sigma_1 \mathbb{I}(u \leq 0.5) + \sigma_2 \mathbb{I}(u > 0.5)$.
- (v) **Abrupt variance break:** $\sigma(u) = \sigma_1 \mathbb{I}(u \leq 0.5) + \sigma_2 \mathbb{I}(u > 0.5)$,

where for the case (iii) we consider $\sigma_1 = 0.5, \sigma_2 = 1$ simulating a regime change variance. where for case (iii) we set $\sigma_1 = 1.0$ and $\sigma_2 = 2.0$, generating an abrupt regime change with a variance jump of 1.0.

Target Parameter and Hypotheses Because the intercept $\mu(u)$ is filtered through the autoregressive dynamics, the true local expected value at time u is given by $m(u) = \mu(u)/(1 - \sum_{j=1}^p a_j)$. Our objective is to estimate the integrated location parameter over the full time horizon, $F(1) = \int_0^1 m(u)du$.

We evaluate the estimators under the null hypothesis of structural stability and two alternatives featuring non-stationary location shifts:

$$\begin{aligned} \mathcal{H}_0 : \quad & \mu_0(u) = 0 \\ \mathcal{H}_1 : \quad & \mu_1(u) = \begin{cases} 0 & \text{for } u \leq 0.5 \\ \delta & \text{for } u > 0.5 \end{cases} \quad (\text{Abrupt Shift}) \\ \mathcal{H}_2 : \quad & \mu_2(u) = \delta \cdot u \quad (\text{Gradual Shift}) \end{aligned}$$

where the shift magnitude for this simulation is $\delta = 0.5$. We estimate the local robust location using the Welesch score function $\psi_c(x)$, solving the implicit estimating equation $\mathbb{E}[\psi_c(X_{t,n} - \theta_{t,n})] = 0$. We compare the performance of the Plug-in Estimator $\tilde{\Theta}_n(1)$ against our Linearized Estimator $\hat{\Theta}_n^{\text{lin}}(1)$. Eventually, we evaluated empirical size and power for the CUSUM test based on the Linearized Estimator.

Contamination Scenarios To highlight the robustness of the linearization procedure and change-point tests under atypical observations, we evaluate the DGP across four contamination regimes: Clean (uncontaminated) and three distinct outlier generating mechanisms:

- (i) **Scenario 1 (Additive Outliers – AO):** We contaminate the time series directly with independent additive outliers following $X_{t,n}^{\text{obs}} = X_{t,n} + I_t^\varepsilon(-1)^\xi K$.

- (ii) **Scenario 2 (Innovation Outliers – IO):** We inject shocks directly into the structural innovations of Equation (12), $\epsilon_t^{\text{IO}} = \epsilon_t + I_t^\varepsilon(-1)^\xi K$, causing outlier effects to propagate dynamically via the autoregressive filter.
- (iii) **Scenario 3 (Replacement Outliers – RO):** We model localized bursts or replacement spikes where contaminated values overwrite the underlying process, $X_{t,n}^{\text{obs}} = (1 - S_t)X_{t,n} + S_t(-1)^\xi K$. Here, $S_t \in \{0, 1\}$ is a 2-state stationary Markov chain with recovery rate $\kappa = 0.50$ and transition probabilities $P(S_t = 1 \mid S_{t-1} = 0) = \varepsilon\kappa/(1-\varepsilon)$ and $P(S_t = 1 \mid S_{t-1} = 1) = 1 - \kappa$, yielding stationary marginal contamination frequency $\mathbb{E}[S_t] = \varepsilon$.

Here, $I_t^\varepsilon \sim \text{Bernoulli}(\varepsilon)$ denotes an outlier indicator with occurrence probability ε , $\xi \sim \text{Bernoulli}(0.5)$ determines the shock sign, and the magnitude is fixed at $K = 10$. An illustrative realization of sample size $n = 200$ under an abrupt shift $\mathcal{H}_1$ is depicted in Figure 1 across all four regimes.

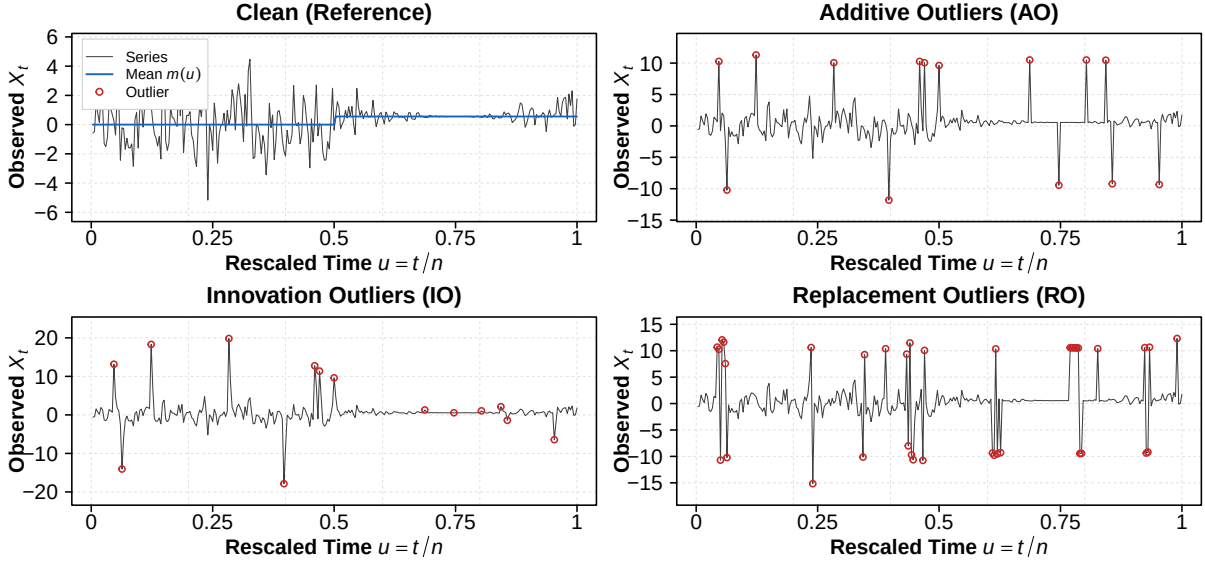


Figure 1: **Simulated trajectories from the location DGP across contamination scenarios under an abrupt structural break ($\mathcal{H}_1$).** The clean baseline (top left) follows an ARMA(2,1) model with time-varying scale $\sigma(u) = \exp(u)$ and an abrupt jump $\delta = 0.50$ at $u = 0.50$; the blue solid line indicates the true theoretical expectation $m(u) = \mu(u)/(1 - a_1 - a_2)$. The remaining panels depict the three contamination mechanisms with contamination probability $\varepsilon = 0.05$ and shock magnitude $K = 10$: Additive Outliers (AO, top right), Innovation Outliers (IO, bottom left), where innovation shocks propagate through autoregressive filtering, and Replacement Outliers (RO, bottom right), where a 2-state Markov chain generates persistent contamination clusters. Red circles mark the observations affected by contamination.

Bandwidth Selection Strategy The practical implementation of our change-point framework requires specifying the local estimation bandwidth M . In this simulation, we use M that minimizes the system of equations on an unseen future observation at lag L_n . Formally, the optimal bandwidth is selected by solving:

$$M^{\text{opt}} = \arg \min_{M \in \mathcal{M}_n} \Lambda(M), \quad \text{where} \quad \Lambda(M) = \sum_{t=k}^{N-L_n} \left\| H \left(X_{t+L_n,n}, \hat{\theta}_t(M) \right) \right\|^2 \quad (13)$$

where $\mathcal{M}_n = \{\lfloor n^{0.45} \rfloor, \lfloor n^{0.65} \rfloor\}$ defines the admissible search space ensuring asymptotic consistency, and $L_n = \lfloor \frac{1}{10}(\log n)^2 \rfloor$ denotes the decoupling lag. Although an automatic selection procedure for L_n would be desirable, we adopt this choice following Mies (2023), where it demonstrated the best overall empirical performance. Finally, we set the window size for variance estimation to $b_n = L_n$.

Simulation Results We divide our numerical evaluation into two complementary analyses: first, we examine the finite-sample estimation accuracy (Mean Absolute Error and empirical Bias) of the proposed linearized recursive estimator $\hat{\Theta}_n^{\text{lin}}(1)$ relative to the standard non-linearized plug-in estimator $\hat{\Theta}_n(1)$; second, we investigate the empirical size and power of our proposed CUSUM test against leading robust change-point detection procedures.

Point Estimation: Bias Correction and Robustness Table 1 reports the Mean Absolute Error (MAE) and empirical Bias for $\hat{\Theta}_n^{\text{lin}}(1)$ and $\tilde{\Theta}_n(1)$ across sample sizes $n \in \{100, 500, 1000, 5000\}$ under variance scenario (i) ($\sigma(u) = \exp(u/2)$). Under the null hypothesis of structural stability ($\mathcal{H}_0$), both estimators perform nearly identically, exhibiting negligible bias (< 0.002 in magnitude for $n \geq 500$) and monotonically decaying MAE of order $O(n^{-1/2})$.

Under the alternative hypotheses ($\mathcal{H}_1$ and $\mathcal{H}_2$), however, their finite-sample behaviors diverge significantly. Because standard rolling-window M-estimation suffers from an intrinsic tracking delay across structural shifts, the non-linearized estimator $\tilde{\Theta}_n(1)$ accumulates severe lag bias. At $n = 100$, $\tilde{\Theta}_n(1)$ exhibits a bias of -0.047 under Clean Gaussian data for $\mathcal{H}_1$, remaining as high as -0.009 even at $n = 5000$. In sharp contrast, the linearized estimator $\hat{\Theta}_n^{\text{lin}}(1)$ analytically removes this first-order bias, reducing it to -0.008 at $n = 100$ and 0.000 at $n = 5000$, while consistently lowering the MAE across all sample sizes.

Importantly, this bias correction is fully robust to heavy-tailed innovations (t_3) and outlier contamination regimes. Under Innovation Outliers (IO), where shocks propagate through the autoregressive dynamics and create localized bursts of outliers, standard estimation is vulnerable to persistent tracking distortion; here, $\hat{\Theta}_n^{\text{lin}}(1)$ achieves a major reduction in both bias (e.g., at $n = 100$, -0.003 vs. -0.058 for t_3) and MAE (0.243 vs. 0.274), illustrating the practical benefit of combining the smooth Welsh loss with analytical linearization.

Table 1: Mean Absolute Error (MAE) and Bias of $\hat{\Theta}_n^{\text{lin}}(1)$ and $\tilde{\Theta}_n(1)$ as estimators of $\int_0^1 m(u)du$ under structural breaks and data contamination. Values based on Monte Carlo replications.

Hypothesis	Scenario	Innovations	Metric	n = 100		n = 500		n = 1000		n = 5000	
				$\tilde{\Theta}_n$	$\hat{\Theta}_n^{\text{lin}}$	$\tilde{\Theta}_n$	$\hat{\Theta}_n^{\text{lin}}$	$\tilde{\Theta}_n$	$\hat{\Theta}_n^{\text{lin}}$	$\tilde{\Theta}_n$	$\hat{\Theta}_n^{\text{lin}}$
$\mathcal{H}_0$ (No Shift)	Clean	Gaussian	MAE	0.127	0.134	0.060	0.063	0.042	0.043	0.019	0.019
			Bias	-0.008	-0.006	-0.001	-0.001	0.001	0.001	-0.001	-0.000
		t_3	MAE	0.164	0.170	0.073	0.077	0.056	0.058	0.025	0.025
			Bias	0.019	0.015	-0.000	0.001	-0.002	-0.002	0.000	0.000
	AO	Gaussian	MAE	0.144	0.150	0.061	0.064	0.044	0.046	0.021	0.021
			Bias	-0.004	-0.002	0.001	0.001	-0.002	-0.002	0.000	0.000
		t_3	MAE	0.194	0.197	0.084	0.085	0.063	0.063	0.027	0.028
			Bias	0.002	-0.000	-0.000	-0.002	-0.001	-0.001	-0.002	-0.002
	IO	Gaussian	MAE	0.213	0.192	0.078	0.076	0.054	0.054	0.023	0.023
			Bias	-0.002	0.001	-0.002	-0.000	-0.002	-0.003	0.000	0.000
		t_3	MAE	0.274	0.243	0.111	0.106	0.077	0.075	0.030	0.031
			Bias	0.021	0.015	-0.007	-0.008	-0.000	0.001	0.000	0.000
$\mathcal{H}_1$ (Abrupt)	Clean	Gaussian	MAE	0.140	0.143	0.066	0.064	0.047	0.044	0.022	0.021
			Bias	-0.047	-0.008	-0.028	-0.002	-0.023	-0.002	-0.009	0.000
		t_3	MAE	0.176	0.171	0.081	0.078	0.061	0.059	0.027	0.026
			Bias	-0.057	-0.014	-0.031	0.000	-0.027	-0.003	-0.013	-0.001
	AO	Gaussian	MAE	0.153	0.148	0.066	0.063	0.050	0.046	0.023	0.022
			Bias	-0.061	-0.014	-0.031	-0.001	-0.024	-0.002	-0.010	0.001
		t_3	MAE	0.210	0.207	0.089	0.084	0.064	0.061	0.031	0.029
			Bias	-0.061	-0.008	-0.034	0.001	-0.022	0.003	-0.014	-0.000
	IO	Gaussian	MAE	0.216	0.181	0.082	0.077	0.061	0.055	0.027	0.025
			Bias	-0.067	-0.013	-0.033	0.004	-0.031	-0.004	-0.015	-0.001
		t_3	MAE	0.283	0.243	0.113	0.110	0.077	0.072	0.035	0.033
			Bias	-0.058	-0.003	-0.032	0.006	-0.028	0.002	-0.015	0.001
$\mathcal{H}_2$ (Gradual)	Clean	Gaussian	MAE	0.139	0.143	0.065	0.061	0.047	0.044	0.022	0.020
			Bias	-0.047	-0.014	-0.032	-0.006	-0.023	-0.002	-0.012	-0.002
		t_3	MAE	0.177	0.172	0.089	0.083	0.060	0.058	0.027	0.026
			Bias	-0.054	-0.020	-0.042	-0.014	-0.027	-0.003	-0.013	0.001
	AO	Gaussian	MAE	0.151	0.151	0.073	0.069	0.049	0.046	0.024	0.021
			Bias	-0.059	-0.019	-0.034	-0.005	-0.024	-0.001	-0.012	-0.001
		t_3	MAE	0.199	0.190	0.092	0.085	0.067	0.064	0.031	0.028
			Bias	-0.072	-0.026	-0.039	-0.007	-0.027	-0.001	-0.016	-0.002
	IO	Gaussian	MAE	0.222	0.187	0.083	0.078	0.060	0.055	0.028	0.024
			Bias	-0.066	-0.022	-0.035	-0.003	-0.029	-0.002	-0.017	-0.002
		t_3	MAE	0.275	0.238	0.117	0.105	0.080	0.075	0.036	0.033
			Bias	-0.066	-0.026	-0.046	-0.011	-0.032	-0.005	-0.019	-0.003

Change-Point Testing: Size Control and Heteroskedasticity Invariance We next benchmark our proposed linearized Welsh CUSUM test (**L**) against four leading robust and heteroskedasticity-aware change-point procedures: (i) the Huber loss M -estimator test (**H**) of Hušková and Marušiačková (2012); (ii) the non-parametric Wilcoxon test (**W**) of Dehling et al. (2015); (iii) the two-sample Hodges–Lehmann test (**T**) of Dehling et al. (2020); and (iv) the self-normalized Gini-type test (**S**) of Schmidt (2021), explicitly designed to test for mean breaks under unconditional heteroskedasticity using block-based subsampling. Figure 2 displays the empirical rejection frequencies across sample sizes $n \in \{200, 500, 700\}$ under the nominal significance level $\alpha = 0.05$.

The empirical size results under $\mathcal{H}_0$ (first column of Figure 2) reveal a sharp demarcation between robustness to outlier shocks and robustness to non-stationary volatility. Under smooth trending variance (case (i), blue circles), all tests maintain reasonable size control close to the nominal 5% level. However, when subjected to cyclic heteroskedasticity (case (ii), green circles) or an abrupt variance break (case (iii), red circles), classical robust benchmark methods exhibit severe distortions:

- The two-sample Hodges–Lehmann test (**T**) experiences a catastrophic size collapse under cyclic variance, falsely rejecting the null in over 60% of replications (reaching 73.8% at $n = 700$ for clean Gaussian series). These points exceed the 0.25 plotting ceiling and are off-scale. Under the variance break (case (iii)), its false rejection rate also inflates to 11.6% – 14.2%. Because two-sample rank statistics evaluate pairwise differences $X_j - X_i$ between sub-samples, asymmetric dispersion between the first and second halves of the series fundamentally distorts the rank distribution, which the test misinterprets as a location shift.
- The Wilcoxon test (**W**) similarly shows marked size inflation under cyclic variance, rejecting up to 10.4% under the null.
- The Huberized CUSUM (**H**) displays greater resilience than rank-based methods, keeping empirical size between 4.6% and 9.2%. Because Huber’s score $\psi_c(\cdot)$ is an odd function, $\mathbb{E}[\psi_c(X_t/\sigma(u))] = 0$ holds pointwise for symmetric distributions, preventing spurious mean drift. Nevertheless, because **H** studentizes using a single global long-run variance estimator that assumes stationarity, cyclic volatility still causes non-negligible size inflation.
- The Gini-based test (**S**) of Schmidt (2021) successfully accommodates unconditional heteroskedasticity through its block-based subsampling variance estimator, maintaining stable size under smooth trending and cyclic variance. However, because it operates directly on raw block averages without robust bounded score functions, its finite-sample calibration degrades in the presence of heavy-tailed innovations and clustered replacement outliers.
- In sharp contrast, our proposed linearized test (**L**) is the **only method in the study that achieves complete invariance to heteroskedasticity**. Across all three variance scenarios, its average empirical size remains tightly anchored around the nominal level: 5.7% for case (i), 5.9% for case (ii), and 5.8% for case (iii). By integrating local non-parametric variance estimation with boundary decoupling at lag L_n , **L** standardizes the partial sums dynamically, guaranteeing that volatility shifts are never conflated with breaks in location.

Power Evaluation under Abrupt and Gradual Shifts Columns 2 and 3 of Figure 2 report empirical power under abrupt ($\mathcal{H}_1$) and gradual ($\mathcal{H}_2$) location changes with shift magnitude $\delta = 0.5$. In comparing power, we focus on baseline environments where tests maintain valid size control (case (i)), since the high rejection rates of test **T** under cyclic heteroskedasticity are spurious artifacts of its 63% false alarm rate.

Under $\mathcal{H}_1$ (abrupt break at $u = 0.5$), detection power increases rapidly toward 1.0 as n grows from 200 to 700 ($0.434 \rightarrow 0.832$ for **L**). In case (i), the power of our proposed method (0.598) closely matches that of the Huberized CUSUM (0.592) and rank-based tests (0.626 for **W**, 0.634 for **T**), demonstrating that the heteroskedasticity robustness of **L** incurs no power penalty under well-behaved stationary volatility.

Under $\mathcal{H}_2$ (gradual linear change), the location change accumulates smoothly throughout the horizon, making the break more difficult to identify locally. Nonetheless, power increases monotonically with n across all configurations. Furthermore, under Innovation Outliers (IO) and Replacement Outliers (RO), where shocks propagate or persistently overwrite observations, the smooth exponential downweighting of the Welsh loss $\psi(u) = u \exp(-(u/c)^2)$ prevents contamination bursts from destabilizing the CUSUM path, yielding superior stability and power progression over competing rank and block-based tests.

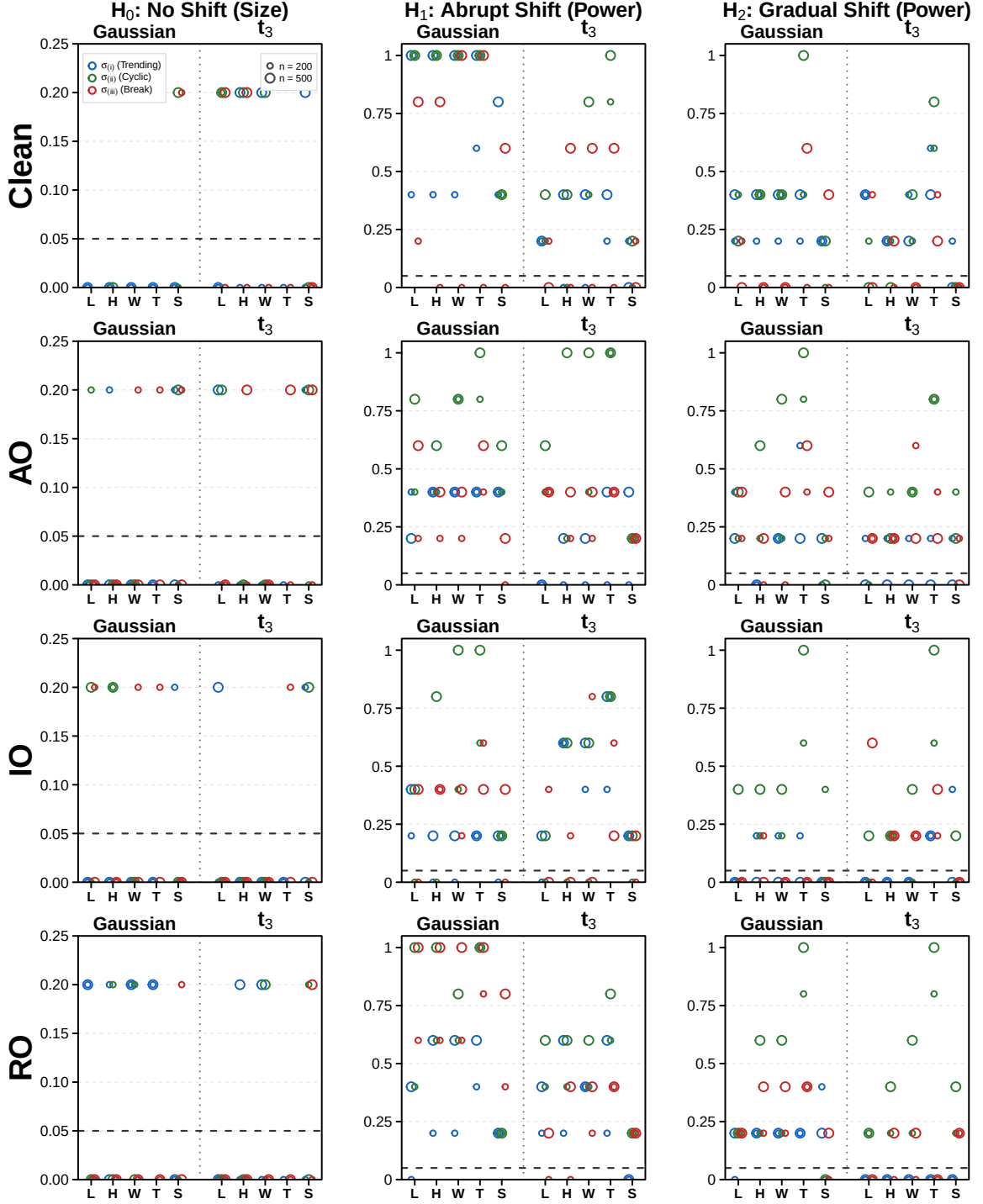


Figure 2: **Empirical rejection rates for robust location change-point tests under ARMA(2,1) dependence.** Evaluated under H_0 (size), H_1 (abrupt break at $u = 0.5$), and H_2 (gradual linear change) with shift magnitude $\delta = 0.5$ at nominal level $\alpha = 0.05$ (dashed line). Rows represent Clean, AO, IO, and RO with $\varepsilon = 0.05$ scenarios for Gaussian and Student- t_3 errors. Methods: **L** (Proposed Linearized Welsch CUSUM), **H** (Huberized CUSUM; Hušková and Marušiačková (2012)), **W** (Wilcoxon test; Dehling et al. (2015)), **T** (Hodges-Lehmann test; Dehling et al. (2020)), and **S** (Schmidt’s Gini test; Schmidt (2021)). Circle colors represent variance scenarios: smooth trending (blue), cyclic (green), and abrupt break (red). Open circle markers scale with sample size $n \in \{200, 500, 700\}$.

5.2 Time-Varying Robust M-Regression

In this section, we examine the finite-sample performance of the proposed CUSUM test for the robust time-varying regression model introduced in Example 2.5. Let $D_{t,n} = (X_{t,n}^\top, Y_{t,n})^\top \in \mathbb{R}^{d+1}$, for $t = 1, \dots, n$, be

a triangular array of observations governed by the causal locally stationary data-generating mechanism

$$Y_{t,n} = X_{t,n}^\top \beta \left(\frac{t}{n} \right) + \nu_{t,n}, \quad t = 1, \dots, n, \quad (14)$$

where $X_{t,n} \in \mathbb{R}^d$ is a vector of locally stationary regressors, $\beta(u) = (\beta_1(u), \dots, \beta_d(u))^\top \in \mathcal{C}^2([0, 1], \mathbb{R}^d)$ represents the vector of time-varying regression coefficients, and $\nu_{t,n}$ is a process satisfying the moment condition $\mathbb{E}[\psi(Y_{t,n} - X_{t,n}^\top \beta(t/n)) \mid \mathcal{F}_{t-1}, X_{t,n}] = 0$ almost surely, with $\psi(\cdot)$ being a robust score function $\in \mathcal{C}^2([0, 1])$.

We are interested in testing whether the regression parameters are structurally stable over the observation horizon, let $\mathbf{C} \in \mathbb{R}^{s \times d}$ be the selection matrix with rank $\{s \leq d\}$ we want to test:

$$\mathcal{H}_0 : \mathbf{C}\beta(u) \equiv \beta_C \quad \text{for all } u \in [0, 1] \quad \text{vs.} \quad \mathcal{H}_1 : \beta(u) \neq \beta_C \quad \text{for some } u \in [0, 1].$$

Simulation Data Generating Processes To assess both size control under non-stationary nuisance dynamics and power against structural changes, we generate observations under two distinct specifications.

We first introduce the building blocks of the data-generating mechanisms. Let $u = t/n \in [0, 1]$ denote the normalized time index. We define the bivariate covariate process $X_{t,n} = (x_{t,n}^1, x_{t,n}^2)^\top \in \mathbb{R}^2$, where each component $x_{t,n}^i$ is represented as a time-varying linear filter of the past innovation shocks $\{\epsilon_{i,t}\}_{t=-\infty}^\infty$ for $i \in \{1, 2\}$:

$$x_{t,n}^i = \sum_{j=0}^{\infty} \left[\phi_i \left(\frac{t}{n} \right) \right]^j \epsilon_{i,t-j}, \quad i \in \{1, 2\},$$

where the geometric decay functions satisfy $\sup_{u \in [0, 1]} |\phi_i(u)| < 1$ to guarantee short-range dependence and uniform ergodicity. Specifically, we calibrate the time-varying memory via $\phi_1(u) = 0.5 - 0.5u$ and $\phi_2(u) = 0.25 + 0.5u$. The innovation sequences $\{\epsilon_{1,t}\}$ and $\{\epsilon_{2,t}\}$ consist of independent standard normal random variables.

The baseline error sequence $e_{t,n}$ is generated from innovation shocks $\{\eta_t\}_{t=-\infty}^\infty$:

$$e_{t,n} = \sum_{j=0}^{\infty} \left[\xi \left(\frac{t}{n} \right) \right]^j \eta_{t-j},$$

with time-varying persistence $\xi(u) = 0.5 - (u - 0.5)^2 \in [0.25, 0.5]$. The innovations η_t are drawn either from a standard Gaussian distribution $\mathcal{N}(0, 1)$ or a heavy-tailed Student's t_3 distribution.

We consider the following two models:

(i) **Model I: Linear Model with Conditional Heteroscedasticity (LMCH).**

$$Y_{t,n} = X_{t,n}^\top \beta \left(\frac{t}{n} \right) + \Gamma(X_{t,n}) e_{t,n}, \quad (15)$$

where the conditional scale function induces severe regressor-dependent heteroscedasticity via

$$\Gamma(X_{t,n}) = \sqrt{1 + \|X_{t,n}\|_2^2} = \sqrt{1 + (x_{t,n}^1)^2 + (x_{t,n}^2)^2}.$$

Because $\mathbb{E}[e_{t,n} \mid X_{t,n}] = 0$, the conditional moment condition $\mathbb{E}[\psi(Y_{t,n} - X_{t,n}^\top \beta(t/n)) \mid X_{t,n}] = 0$ holds exactly, while the noise variance $\text{Var}(\nu_{t,n} \mid X_{t,n}) = \Gamma(X_{t,n})^2 \text{Var}(e_{t,n})$ fluctuates substantially with the covariate paths, rigorously challenging the robustness of the variance estimator.

(ii) **Model II: Linear Model with Unconditional Heteroscedasticity (LMUH).**

$$Y_{t,n} = X_{t,n}^\top \beta \left(\frac{t}{n} \right) + \sigma \left(\frac{t}{n} \right) e_{t,n}, \quad (16)$$

Where the deterministic volatility trend is given by $\sigma(u) = 0.5 + |\sin(2\pi u)|$, and the innovation process $e_{t,n}$ is conditionally independent of $X_{t,n}$.

Target Parameter and Hypotheses To evaluate both the size control under the null hypothesis and the sensitivity of the proposed test against structural instability, we test the constancy of the regression parameters against two canonical departure mechanisms parameterized by a perturbation scalar $\delta \geq 0$. Following the benchmark designs in ?, Mies (2023) and ?, we focus on an isolated regime break and a continuous macroscopic trend:

$$\begin{aligned}\mathcal{H}_0 : \quad & \beta(u) = \beta_0 \\ \mathcal{H}_1 : \quad & \beta(u) = \beta_0 + \delta \mathbf{v} \mathbf{1}_{\{u > 0.5\}}, \quad u \in [0, 1] \quad (\text{Abrupt Shift}) \\ \mathcal{H}_2 : \quad & \beta(u) = \beta_0 + \delta \mathbf{v} u, \quad u \in [0, 1] \quad (\text{Gradual Shift})\end{aligned}$$

where $\mathbf{1}_{\{\cdot\}}$ is the indicator function and $\mathbf{v} \in \mathbb{R}^d$ is a directional contrast vector that dictates which coefficient undergoes structural transition. For Model I (LMHC), we set $\beta_0 = (1.0, 2.0)^\top$ and perturb the primary exogenous covariate by choosing $\mathbf{v} = (1, 0)^\top$. For Model II (LMAT), where the stacked baseline parameter is $\beta_0 = (0.4, 1.0, 0.5)^\top$, we perturb the first exogenous predictor using $\mathbf{v} = (0, 1, 0)^\top$, keeping the autoregressive parameter fixed at $\beta_{\text{AR}} = 0.4$ to guarantee causal stability across the entire horizon.

Contamination Scenarios To investigate the finite-sample robustness of our proposed CUSUM test against both isolated anomalies and structural contamination, we subject the Data Generating Processes (DGPs) in (15) and (16) to three distinct outlier regimes. Throughout, let $\varepsilon = 0.05$ denote the nominal contamination rate, $I_t^\varepsilon \stackrel{\text{i.i.d.}}{\sim} \text{Bernoulli}(\varepsilon)$ be the contamination indicator, $K = 10$ represent the shock magnitude, and $\xi_t \stackrel{\text{i.i.d.}}{\sim} \text{Bernoulli}(0.5)$ be independent sign-assignment variables such that $(-1)^{\xi_t} \in \{-1, 1\}$.

- (i) **Scenario 1: Additive Outliers (AO).** We generate additive outliers in the predictor response $Y_{t,n}$ via:

$$Y_{t,n}^{\text{AO}} = Y_{t,n} + I_t^\varepsilon (-1)^{\xi_t} K,$$

Because the true latent path is uncorrupted, the shock at time t does not propagate into future time periods.

- (ii) **Scenario 2: Innovation Outliers (IO).** We contaminate the baseline white noise η_t driving $e_{t,n}$ prior to filtering:

$$\eta_t^{\text{IO}} = \eta_t + I_t^\varepsilon (-1)^{\xi_t} K,$$

The contaminated error process $e_{t,n}^{\text{IO}} = \sum_{j=0}^\infty [\xi(t/n)]^j \eta_{t-j}^{\text{IO}}$ dynamically inherits the memory of past contamination shocks, producing localized outlier clusters that challenge standard non-robust variance estimation.

- (iii) **Scenario 3: Replacement Outliers (RO).** While AO and IO have already been discussed in Section 5.1, here we introduce an additional contamination scenario based on replacement. Under this scheme, the observed response variable $Y_{t,n}^{\text{RO}}$ is generated by completely overwriting the latent clean process $Y_{t,n}$ during contaminated periods:

$$Y_{t,n}^{\text{RO}} = (1 - \delta_t) Y_{t,n} + \delta_t Y_{t,n}^*,$$

where $Y_{t,n}^*$ denotes an independent contaminating replacement process and $\delta_t \in \{0, 1\}$ is a binary switching indicator with marginal contamination probability $\mathbb{P}(\delta_t = 1) = \varepsilon$. Notice that standard additive contamination can be recovered as a degenerate special case where $Y_{t,n}^* = Y_{t,n} + (-1)^{\xi_t} K$.

Crucially, the replacement framework allows us to simulate more realistic and challenging patch (or burst) outliers by modeling δ_t via a two-state Markov chain $S_t \in \{0, 1\}$. Let $S_t = 1$ denote the active replacement regime and $S_t = 0$ denote the clean baseline regime. To calibrate the stationary marginal contamination probability to $\mathbb{P}(S_t = 1) = \varepsilon$ and the recovery probability to $\mathbb{P}(S_t = 0 \mid S_{t-1} = 1) = \kappa \in (0, 1)$, detailed balance requires the contamination onset probability to satisfy:

$$\mathbb{P}(S_t = 1 \mid S_{t-1} = 0) = \frac{\mathbb{P}(S_{t-1} = 1) \mathbb{P}(S_t = 0 \mid S_{t-1} = 1)}{\mathbb{P}(S_{t-1} = 0)} = \frac{\kappa \varepsilon}{1 - \varepsilon}.$$

During contaminated periods ($S_t = 1$), the clean observation is overwritten using

$$\begin{aligned}Y_{t,n}^* &= X_{t,n}^\top \left(\gamma \beta \left(\frac{t}{n} \right) \right) + \Gamma(X_{t,n}) e_{t,n} && \text{For Model I (LMHC)} \\ Y_{t,n}^* &= W_{t,n}^\top \left(\gamma \beta \left(\frac{t}{n} \right) \right) + \sigma \left(\frac{t}{n} \right) e_{t,n} && \text{For Model II (LMAT)}\end{aligned}$$

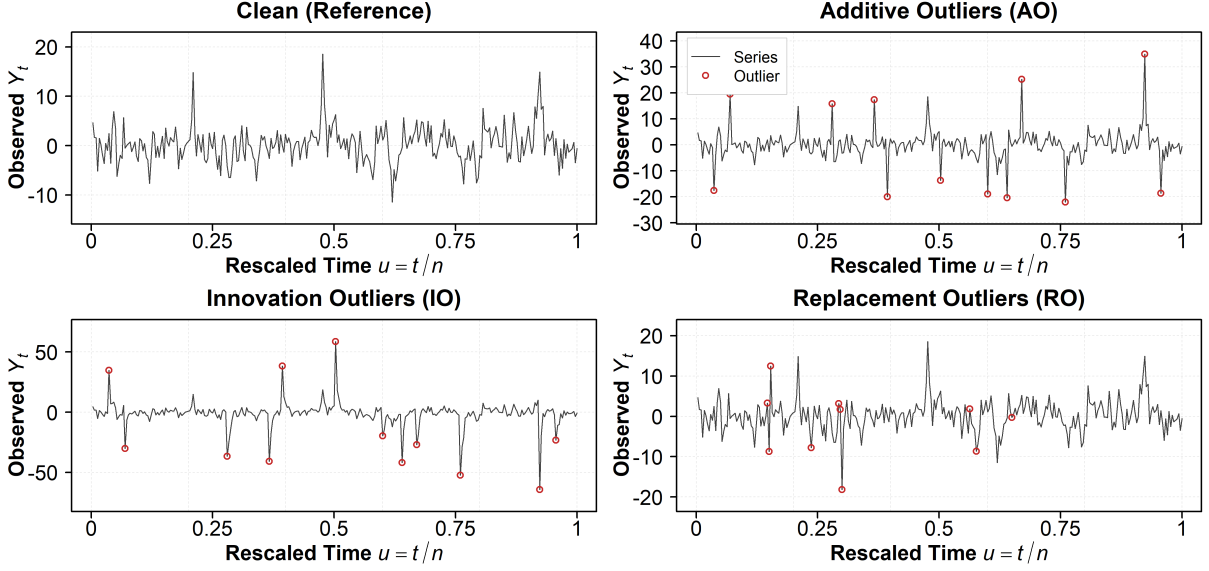


Figure 3: Caption

This design induces localized patches of outliers that test the resistance of the local M-estimating function when the location is not random as in the AO scenario.

Implementation Details and Bandwidth Selection We evaluate the finite-sample performance of the proposed linearized recursive CUSUM test across $N_{\text{sim}} = 1,000$ independent Monte Carlo replications for sample sizes $n \in \{500, 1,000, 5,000\}$ under a nominal significance level $\alpha = 0.05$. The multiplier bootstrap approximations are conducted using $B = 100$ bootstrap draws per replication. For the alternative departure mechanisms under $\mathcal{H}_1$ and $\mathcal{H}_2$, the perturbation magnitude is set to $\delta = \{0, 0.25, 0.5, 0.75, 1, 1.25\}$

Results We compare the performance of the CUSUM test using the proposed linearized M-estimator with both Welsch and squared loss functions. The squared loss, $\rho(r) = \frac{1}{2}r^2$, corresponds to the non-robust CUSUM procedure introduced in Mies (2023). Throughout, we fix the Welsch tuning parameter to $c = 2.985$ to guarantee 95% asymptotic relative efficiency under Gaussian innovations, whereas for the selection of M , L_n , and b_n , we follow the smae procedure used in Section 5.1. Table 2 summarizes the empirical rejection frequencies under the null hypothesis $\mathcal{H}_0$ at the nominal level $\alpha = 0.05$. In general, the test based on the L_2 loss is notably conservative, which is consistent with the findings in Mies (2023). Interestingly, even in the clean setting with Gaussian innovations, the Welsch loss maintains an empirical size closer to the nominal level, despite a slight upward distortion of roughly 0.02, whereas the L_2 loss yields an empirical size of only 0.011. Moreover, under the IO scenario, both estimators become more conservative. We next analyze empirical power. Figures ?? and 5 display the power curves for standard normal and t_3 innovations, respectively, under both alternative specifications. In the clean scenario across both figures, both loss functions exhibit comparable performance; however, the Welsch loss attains higher power earlier, detecting smaller jump sizes δ . In contaminated settings, the performance of the L_2 loss deteriorates substantially—most noticeably under IO contamination, where the test suffers from low power even when $n = 5000$. While these contaminated setups naturally favor robust methods, the proposed M-estimation framework systematically enhances detection performance across the board.

5.3 SEDCD

To assess the finite-sample performance of the proposed linearized CUSUM test for detecting structural breaks in extreme dependence, we conduct a comprehensive Monte Carlo simulation study. Our experimental design evaluates both the empirical size (under the null hypothesis of a constant extreme dependence structure) and the empirical power (under the alternative hypothesis of a changing dependence structure). Crucially, we test the robustness of the procedure to non-stationary nuisance parameters by introducing structural changes exclusively in the marginal distributions.

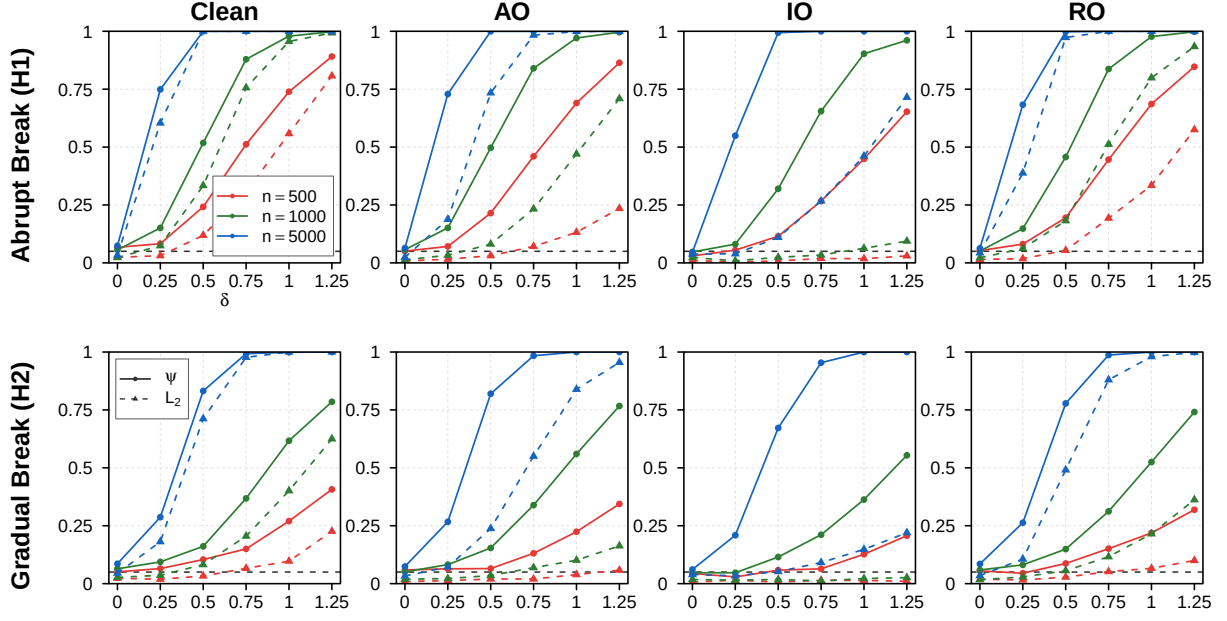


Figure 4: **Empirical power curves for robust regression change-point tests under Gaussian noise.** Rejection rates against shift magnitude $\delta \in [0, 1.25]$ under Abrupt (H_1 , top) and Gradual (H_2 , bottom) structural breaks across Clean, AO, IO, and RO contamination scenarios ($\varepsilon = 0.05$). Solid lines (ψ) show the proposed Linearized Robust M-test with Welsch loss; dashed lines (L_2) show the classical non-robust CUSUM test. Colors indicate $n \in \{500, 1000, 5000\}$ ($N^{sim} = 1,000$ replications, $B = 100$ bootstrap sample, nominal level $\alpha = 0.05$).

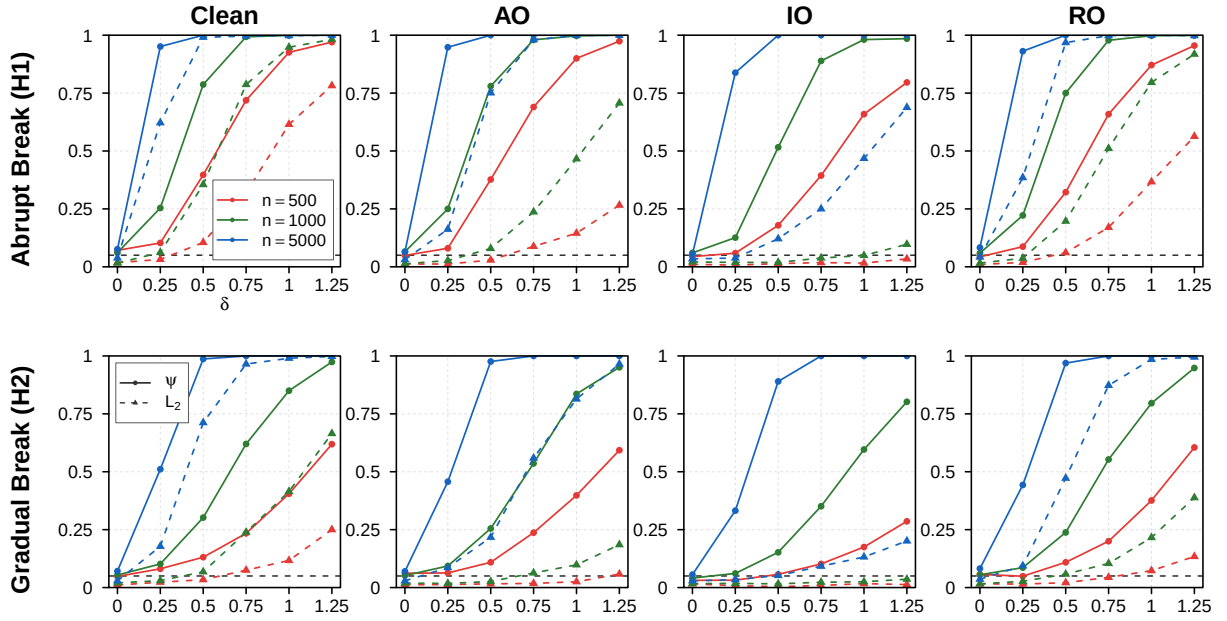


Figure 5: **Empirical power curves for robust regression change-point tests under t_3 noise.** Rejection rates against shift magnitude $\delta \in [0, 1.25]$ under Abrupt (H_1 , top) and Gradual (H_2 , bottom) structural breaks across Clean, AO, IO, and RO contamination scenarios ($\varepsilon = 0.05$). Solid lines (ψ) show the proposed Linearized Robust M-test with Welsch loss; dashed lines (L_2) show the classical non-robust CUSUM test. Colors indicate $n \in \{500, 1000, 5000\}$ ($N^{sim} = 1,000$ replications, $B = 100$ bootstrap sample, nominal level $\alpha = 0.05$).

Table 2: Empirical size rejection frequencies under the null hypothesis $\mathcal{H}_0$ (nominal level $\alpha = 0.05$) across $N_{\text{sim}} = 1,000$ Monte Carlo replications for sample sizes $n \in \{500, 1,000, 5,000\}$ under Model I (LMHC).

Model	Noise	Regime	ε	Linearized M-Test			Linearized L_2 Test		
				$n = 500$	$n = 1000$	$n = 5000$	$n = 500$	$n = 1000$	$n = 5000$
Model I (LMHC)	$\mathcal{N}(0, 1)$	Clean	0.00	0.079	0.075	0.067	0.012	0.012	0.011
		AO	0.05	0.067	0.074	0.065	0.004	0.008	0.011
		IO	0.05	0.020	0.021	0.028	0.007	0.009	0.018
		RO	0.05	0.077	0.054	0.071	0.016	0.010	0.028
	t_3	Clean	0.00	0.061	0.062	0.068	0.013	0.011	0.025
		AO	0.05	0.047	0.052	0.057	0.002	0.006	0.014
		IO	0.05	0.014	0.012	0.026	0.007	0.009	0.024
		RO	0.05	0.050	0.058	0.058	0.010	0.019	0.022

We consider multivariate ($d = 2, d = 5$) time series with sample lengths of 2000. To approximate the asymptotic critical values, we employ the multiplier bootstrap scheme with $B = 100$ replications for each Monte Carlo iteration $M = 1000$.

To ensure our simulations capture extreme structural co-movements in the lower tail of the multivariate distribution, we utilize two distinct parametric copula families:

Clayton Copula. For a d -dimensional vector of uniform marginals $u_1, u_2, \dots, u_d \in [0, 1]^d$, the multivariate Clayton copula is defined as:

$$C_\theta^d(u_1, \dots, u_d) = \left(\sum_{i=1}^d u_i^{-\theta} - d + 1 \right)^{-1/\theta}, \quad (17)$$

where $\theta \in (0, \infty)$ governs the dependence structure. The Clayton copula exhibits an asymmetric tail dependence. In particular, it possesses positive lower-tail dependence and zero upper-tail dependence. The lower-tail dependence coefficient is given by $\lambda_L = 2^{-1/\theta}$, while $\lambda_U = 0$. Consequently, the Clayton specification is well-suited for modeling asymmetric systemic risk characterized by stronger dependence during extreme joint downturns.

Student-t Copula. For a d -dimensional vector of uniform marginals $u_1, u_2, \dots, u_d \in [0, 1]^d$, the d -dimensional Student-t copula is defined as:

$$C_{\nu, \Sigma}^d(u_1, \dots, u_d) = \mathbf{t}_{\nu, \Sigma}^d(t_\nu^{-1}(u_1), \dots, t_\nu^{-1}(u_d)), \quad (18)$$

where $t_{\nu, \Sigma}^{(d)}$ denotes the cdf of a d -dimensional Student- t distribution with covariance matrix Σ and ν degrees of freedom, and t_ν^{-1} denotes the quantile function of the univariate Student- t distribution. In contrast to the Clayton copula, the Student- t copula exhibits symmetric upper- and lower-tail dependence.

The multivariate DGP $\mathbf{X}_t = (X_{1,t}, \dots, X_{d,t})^\top$ is obtained by transforming each uniform component derived from Equations (17) and (18) through the quantile function of the target distributions, defined as follows:

Student-t Marginals. To introduce heavy tails, a common feature of financial returns, we use the Student- t distribution for the marginals. The cdf is denoted by $F(x; \nu)$, where ν is the degrees of freedom parameter. The baseline observations are generated by applying the inverse cdf (quantile function) to the uniform variates from the copula:

$$X_{i,t} = F^{-1}(U_{i,t}; \nu), \quad \text{for } i = 1, \dots, d \text{ and } t = 1, \dots, n. \quad (19)$$

The degrees of freedom parameter ν directly controls the tail thickness. For $\nu > 2$, the variance is finite, but as ν decreases, the tails become heavier, in order to satisfy Assumption 2.5 we require $\nu > 4$. This setup allows us to test the procedure's performance under controlled degrees of tail extremity.

Table 3: Combined Empirical Size at 10% Significance Level.

DGP	d	tau	n	$0.1(\log(n))^2$	$0.25(\log(n))^2$	$0.5(\log(n))^2$	$0.75(\log(n))^2$	$1(\log(n))^2$
Clayton	2	-2.0	2000	0.020	0.013	0.014	0.018	0.016
Clayton	2	-1.5	2000	0.033	0.027	0.016	0.011	0.011
Clayton	5	-2.0	2000	0.011	0.005	0.002	0.003	0.002
Clayton_Sine	2	-2.0	2000	0.034	0.021	0.020	0.014	0.019
Clayton_Sine	2	-1.5	2000	0.022	0.018	0.011	0.011	0.011
Clayton_Sine	5	-2.0	2000	0.293	0.175	0.060	0.030	0.020
Clayton_Sine	5	-1.5	2000	0.164	0.091	0.032	0.015	0.013
T_Copula	2	-2.0	2000	0.016	0.011	0.011	0.013	0.011
T_Copula	2	-1.5	2000	0.037	0.028	0.024	0.024	0.019
T_Copula	5	-2.0	2000	0.029	0.024	0.016	0.013	0.010
T_Copula	5	-1.5	2000	0.079	0.059	0.045	0.033	0.026
T_Copula_Sine	2	-2.0	2000	0.023	0.018	0.014	0.016	0.019
T_Copula_Sine	2	-1.5	2000	0.029	0.023	0.017	0.020	0.023
T_Copula_Sine	5	-1.5	2000	0.115	0.084	0.059	0.040	0.036

Time-Varying Student-t Marginals. To evaluate the robustness of the testing procedure against non-stationary nuisance parameters, we define a periodic scaling function $c(s)$ on the rescaled time interval $s \in [0, 1]$ as $c(s) = 1 + 0.5 \sin(2\pi s)$.

By applying this time-varying volatility to the quantile function, the non-stationary observations are generated as:

$$X_{i,t} = c\left(\frac{t}{n}\right) \cdot F^{-1}(U_{i,t}; \nu), \quad \text{for } i = 1, \dots, d \text{ and } t = 1, \dots, n. \quad (20)$$

The practical implementation of our change-point framework requires specifying the local estimation bandwidth k . In this simulation we use k that minimizes the system of equations on an unseen future observation at lag L_n . Formally, the optimal bandwidth is selected by solving:

$$k^{\text{OPT}} = \arg \min_{k \in \mathcal{K}_n} \Lambda(k), \quad \text{where } \Lambda(k) = \sum_{t=k}^{N-L_n} \left\| H\left(X_{t+L_n, n}, \hat{\theta}_t(k)\right) \right\|^2 \quad (21)$$

where $\mathcal{K}_n = [N^{0.35}, N^{0.55}]$ defines the admissible search space, ensuring asymptotic consistency and $L_n \in \{[\log n], [n^{0.15}], [n^{0.20}]\}$. This implicit predictive extension ensures that the selected bandwidth directly optimizes the structural stability of the underlying GEE system. Table ?? reports the empirical probability of Type 1 error for different levels of $\tau \in \{-2, -1\}$.

To evaluate the power of the proposed test under the alternative hypothesis, we consider three distinct simulation settings.

Linear Clayton In this setting, the dependence parameter θ of the Clayton copula increases linearly over time, such that $\theta(u) = 2 + 4u$ for $u \in [0, 1]$.

Linear Student's t Similarly, the dispersion matrix $\Sigma(u)$ of the Student's t -copula evolves linearly. Specifically, the time-varying correlation parameter is given by $\rho(u) = \rho_1 u$ for $u \in [0, 1]$ and $\rho_1 \in \{0.25, 0.5, 0.75, 0.95\}$.

Jump Student's t This setting introduces an abrupt structural break at $u = 0.5$. The dispersion matrix $\Sigma(u)$ changes instantaneously according to the correlation parameter:

$$\rho(u) = \begin{cases} \rho_0, & u \leq 0.5 \\ \rho_1, & u > 0.5 \end{cases}$$

where the pre-break correlation is fixed at $\rho_0 = 0$, and the post-break correlation takes values $\rho_1 \in \{0.25, 0.5, 0.75, 0.95\}$. In all proposed settings, we use a marginal t -student with $\nu = 5$

6 Application

7 Discussion

References

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